

The Allegory of the Favela: The Multifaceted Effects of Socioeconomic Mobility

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journals.sagepub.com/home/asq**Leandro S. Pongeluppe**¹ 

Abstract

What are the consequences of upward socioeconomic mobility for disenfranchised individuals? This article examines this question in the context of a business training program offered to residents of Brazilian urban slums, known as “favelas.” The study employs a randomized controlled trial complemented by quantile regressions, field visits, and interviews. The results show that training improves favela dwellers’ economic outcomes, such as by increasing income and participation in entrepreneurship, and some socio-psychological outcomes, such as by improving self-efficacy and optimism. However, these income improvements were accompanied by participants’ enhanced experiences of favela stigma, an adverse socio-psychological outcome related to their residential segregation. Both quantitative and qualitative findings demonstrate the multifaceted nature of socioeconomic mobility, through which favela dwellers who prosper economically become more exposed to prejudice from people living outside favelas. The study illustrates, through the “allegory of the favela,” the bittersweet process of socioeconomic mobility. This abductive research contributes to the literature by showing that while interventions designed to enfranchise individuals may effectively achieve economic inclusion in terms of income gains, they may simultaneously lead participants into discriminatory systems that further stigmatize people based on the same characteristics of their prior exclusion.

Keywords: labor market discrimination, social comparison, inclusion and belonging, randomized controlled trial, Sustainable Development Goals

What are the effects of upward socioeconomic mobility for disenfranchised people? The literature has shown that entrepreneurship can be an opportune career path for disenfranchised individuals facing discrimination when they try to access formal labor markets (Hwang, 2022; Hwang and Phillips, 2023). Thus, interventions such as training programs can most effectively promote

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economic inclusion and prosperity if they focus on developing individuals' skills and capabilities to enter the labor force as entrepreneurs (McKenzie and Woodruff, 2014). A critical consideration is that as economic inclusion occurs, individuals start comparing themselves with a new set of higher-status peers (Merton, 1968; Frank, 1985; Tan and Rider, 2023). These social comparison processes may generate psychological distress as people realize and experience stigmatization, tokenization, and discrimination while conducting their work (Kanter, 2008; Turco, 2010). This study investigates the intended and unintended consequences of a training program dedicated to individuals living in "stigmatized neighborhoods" (Ropert and Di Masso, 2021: 53; Born, 2023). The findings reveal the multifaceted nature of socioeconomic mobility in which individuals who achieve higher income levels also experience the most prejudice due to where they live. This unintended consequence (or "latent function," Merton, 1957: 68) underscores the complexity of socioeconomic inclusion in discriminatory societal systems.

The context of this study is a business training program randomly offered to individuals living in Brazilian urban slums, known as favelas. The study uses a stratified randomized controlled trial to analyze the effects of a training program offered by a non-profit organization, Banco da Providência. The program provides psychological support, technical training, and management coaching to favela dwellers in Rio de Janeiro. The analysis uses two sets of outcomes to evaluate the value generated by the training program: economic outcomes measured by job allocation and income gains and socio-psychological outcomes measured by participants' reports of self-efficacy, optimism, and experiences of social stigma.

The results indicate that the training program had significant positive effects on both economic outcomes and on two of the three socio-psychological outcomes. Yet, the training led participants to experience adverse socio-psychological feelings, namely, increased social stigma related to their residence in favelas. Additional quantitative results demonstrate that the higher the income levels individuals achieved, the more prejudice they experienced. Qualitative evidence suggests that the favela stigma arose because participants became more exposed to social interactions with higher-status, non-favela individuals (potential employers and suppliers), who classified them as untrustworthy and involved in criminal activities. I explain this process through what I call the "allegory of the favela": a contemporary rereading of Plato's allegory of the cave (Bloom, 1968). An allegory is conventionally understood as a story within a story; thus, following Merton (1957: 68), this study demonstrates that interventions primarily targeting economic outcomes, the "manifest" story, may overlook socio-psychological outcomes, the "latent" story, experienced by participants.

The study's findings have important theoretical and empirical implications for the literature. First, I use an abductive approach to theorize the multifaceted nature of socioeconomic mobility. Building on unexpected results of a pre-registered field experiment, I find that socioeconomic inclusion may have intended consequences ("manifest functions") in terms of economic prosperity but also unintended consequences ("latent functions") in terms of socio-psychological distress among participants (Merton, 1957: 68). In showing that income gains go hand in hand with increased experiences of prejudice, this study directly contributes to the literature on labor market discrimination

(Hwang, 2022; Hwang and Phillips, 2023; Rider et al., 2023), social comparison (Merton, 1968; Frank, 1985; Tan and Rider, 2023), and inclusion and feelings of belonging (Kanter, 2008; Turco, 2010; Tewfik, 2022). Second, this research is among the few studies to investigate the causal impact of a training program on economic and socio-psychological outcomes of individuals who are “residentially ‘segregated’” (Massey and Denton, 1988: 283). In emphasizing the importance of both sets of outcomes, the study contributes to the literature by examining individuals holistically and not solely based on their economic production. These themes are relevant to the achievement of the United Nations’ Sustainable Development Goals (SDGs), particularly those related to poverty eradication (SDG #1), inequality reduction (SDG #10), and sustainable cities and communities (SDG #11). Finally, the study aims to call for change in the societal mindset toward disenfranchised people. Such change is essential to breaking the chains of segregation embedded in current socioeconomic systems and allowing individuals to realize their full potential.

ANTECEDENTS IN THE LITERATURE

This section briefly reviews two streams of literature for their theoretical implications: labor market bias and discrimination, as well as social comparison, inclusion, and belonging. The purpose is to identify critical contingencies and relationships stipulated in prior research that will motivate this study’s empirical tests and abductive theorizing.

Labor Market Discrimination, Entrepreneurship, and Training Programs

The literature has shown that due to their condition, disenfranchised individuals often suffer from labor market discrimination (Goldin and Rouse, 2000; Kang et al., 2016; Rider et al., 2023). Recent studies have shown that entrepreneurship can be an opportune career option for these individuals to avoid the discrimination that occurs when they try to access the formal labor market (Hwang, 2022; Hwang and Phillips, 2023). Moreover, entrepreneurship can be an effective career option when institutional frameworks do not provide enough support for individual development (Thébaud, 2015); when individuals’ nationalities and ethnic backgrounds are socially devalued, thereby hindering access to the labor market (Marinoni, 2023); and when individuals lack access to better outside options and need to create new ventures out of necessity (Dencker et al., 2021).

Studies demonstrate that these ventures can be a way to overcome labor market discrimination and contribute to income generation and skill development among disenfranchised people. For instance, Hwang and Phillips (2023) showed how entrepreneurship among formerly incarcerated people in the United States helps these individuals avoid discrimination when searching for formal employment, increases their income levels after being reintroduced into society, and reduces their likelihood of recidivism. In other words, it generates direct socioeconomic benefits for these individuals, their communities, and society at large.

One potential way to increase these positive effects is to foster training among disenfranchised groups to better prepare them to enter the labor

market as new entrepreneurs. Training programs have been widely used to promote local economic development by enhancing the technical and social skills of disenfranchised individuals (Attanasio, Kugler, and Meghir, 2011; de Mel, McKenzie, and Woodruff, 2012; Bulte, Lensink, and Vu, 2017; Carlson and Hager, 2021; Dimitriadis and Koning, 2022). Technical skills refer to professional, managerial, and entrepreneurial abilities, while social skills relate to emotional intelligence, rapport building, communication ability, and other socio-psychological competencies. McKenzie and Woodruff (2014) provide a more extensive literature review of these programs.

Most literature in this area points to the benefits of training programs for improving disenfranchised people's economic outcomes, specifically gains in job allocation, launch and survival of new ventures, and increases in individuals' incomes and ventures' profitability (Kraay and McKenzie, 2014). However, with notable exceptions (e.g., Dimitriadis and Koning, 2022), this literature has traditionally overlooked the effect of training programs on socio-psychological outcomes of disenfranchised people, such as interpersonal skills, self-confidence, optimism, and feelings of inclusion and belonging. To gain more-comprehensive understanding of the consequences of socioeconomic mobility for these groups of people, this study evaluates the effects of a training program on both sets of outcomes.

Social Comparison, Inclusion and Belonging, and Stigmatized Neighborhoods

A fundamental element of socioeconomic mobility, particularly related to socio-psychological outcomes, refers to changes in the peers an individual uses as a reference group for social comparison (Frank, 1985). The literature has documented that as individuals advance in their careers, they tend to compare themselves with higher-status peers (Merton, 1968). This comparison might generate socio-psychological distress since individuals (and organizations) might infer quality through a "socially endogenous calibration" process based on their relative social position as well as peers' perception of them (Tan and Rider, 2023: 509).

The literature on inclusion and feelings of belonging has demonstrated that minority groups and individuals in a "token" position within organizations experience this socio-psychologic distress more severely than do people belonging to majority groups (Kanter, 2008: 6). Scholars aiming to mitigate this distress among disenfranchised individuals typically propose limited interventions, such as nudges and framings, as possible solutions. These simple fixes range from conducting blind auditions for musicians (Goldin and Rouse, 2000) to implementing opt-out policies for managerial position tournaments (He, Kang, and Lacetera, 2021). Despite their insightfulness, these interventions do not address the root causes and the nuances of discriminatory practices in organizations and economic systems. For instance, these interventions may help women musicians to gain access to orchestras previously dominated by men (Goldin and Rouse, 2000), but they do not account for the additional discrimination burden that women newly included as orchestra members might suffer.

The complexities of inclusion and belonging are even more profound. Studies have shown that tokenism is not equal among different groups of disenfranchised individuals. Depending on local cultural contexts and contingencies, tokenism can be harsher on some groups than others (Turco, 2010). For instance, Turco (2010) described how the local cultural context of the leveraged buyout industry in the United States excludes women more than Black men despite both groups being tokenized. According to the author, cultural beliefs about motherhood's impact on work engagement and differences in taste for sports make gender a more relevant status characteristic for exclusion than race in the leveraged buyout industry context. In other words, relative differences in social status affect the socio-psychological outcomes of disenfranchised individuals in their socioeconomic mobility. Moreover, specific characteristics of these individuals determine their inclusion, self-esteem, and sense of belonging (Tewfik, 2022). Furthermore, as Rider and colleagues (2023) demonstrated in their analysis of career advancement among coaches in the National Football League, these organizational processes affect an individual's assignment to work activities (i.e., allocative bias) and promotion possibilities (i.e., valutive bias).

An additional layer of discrimination that remains understudied in this literature refers to the spatial distribution of individuals. The literature on critical geography (Santos, 1978, 1987; Melgaço and Prouse, 2017) shows that physical space has an active role in shaping and constraining socioeconomic opportunities. Examples include research on residential segregation (Massey and Denton, 1988; Massey, 1990; Charles, 2003), redlining policies (Zenou and Boccoard, 2000), stigmatized neighborhoods (Ropert and Di Masso, 2021; Born, 2023), and spatial inequalities (Pongeluppe, 2022). These studies exemplify how a geographic location may represent a symbolic space (Bourdieu, 1991; Wacquant, 2022; Born, 2023) used to arbitrarily categorize the people living in neighborhoods and to stigmatize them and their communities.

This stigmatization process impedes access to educational, cultural, and labor opportunities for the individuals living in these locations. Such individuals are symbolically associated with inappropriate "ghetto behavior" (Smith, 2005: 23), potentially classified as violent, and are thought to be related to criminal activities (Smith, 2005; Larkins, 2015). The stigmatization of individuals based on their place of residence strengthens geographical, economic, and social segregation cycles, which are mutually reinforcing and persistent over time (Massey and Denton, 1988; Massey, 1990; Charles, 2003). This entire process acts similarly to "poverty traps" (Dasgupta, 1997: 5), by which individuals living in stigmatized spaces are stuck in a vicious cycle of disenfranchisement.

FAVELAS AS STIGMATIZED NEIGHBORHOODS AND THE TRAINING PROGRAM DEDICATED TO FAVELA DWELLERS

Brazilian Favelas

"Favela" is the Brazilian Portuguese word for an urban slum. These locations are associated with numerous hurdles to economic development. For example, favelas are characterized by deficient infrastructure, lack of formally defined property rights, recurrent conflicts between gangs and police forces, and the

absence of public services such as health, legal, and education systems (Perlman, 2010; Jovchelovitch and Priego-Hernandez, 2013; Larkins, 2015; Meirelles and Athayde, 2016; Pongeluppe, 2022).

Favelas cover substantial urban areas in Brazil. According to the 2010 Brazilian Census, about 11.4 million people live in favelas, of which 88 percent are located in the 20 largest Brazilian metropolitan regions. Similar settlements also exist outside Brazil. Worldwide, 828 million people live in the equivalent of favelas, and in developing countries one of three urban dwellers lives in an urban slum (UN-Habitat, 2010).

Favelas are the archetype of a stigmatized neighborhood (Ropert and Di Masso, 2021; Born, 2023). Research has shown that favelas are poverty traps (Marx, Stoker, and Suri, 2013) where, once in, individuals cannot escape due to social ostracism and lack of opportunities. The favela stigma has been portrayed in Brazilian cinema in classic movies such as *Five Times Favela* (1962), *City of God* (2002), and *Elite Squad* (2007), in which favela residents are depicted as absorbed by delinquency and predestined by Darwinism to a criminal lifestyle.¹

The negative representation of favela residents reinforces their spatial segregation, social stigmatization, economic disenfranchisement, and lack of self-esteem (Jovchelovitch and Priego-Hernandez, 2013: 200). Moreover, there is evidence that even (or especially) the government has historically contributed to further segregating these communities by making them invisible. For example, during the 2016 Rio de Janeiro Summer Olympics, the city administration erased favelas from the official Olympic map and erected walls alongside roads that crossed favela areas (Arsenault, 2016; Opray, 2016).

The Bolsa Família Program

Given the precarious economic conditions in Brazilian favelas, a significant portion of favela residents are beneficiaries of public welfare (distress) programs, the most prominent of which is Bolsa Família (BF). The program was initially created in 2003 and, by August 2023, had reached 21.14 million families throughout all 5,570 distinct Brazilian municipalities.² Given the program's national reach, the Brazilian federal government called on municipal governments to implement it locally. Municipalities are responsible for enrolling families that have a monthly per capita income less than or equal to R\$85 (USD\$21), which is the eligibility threshold for Bolsa Família cash transfers.³ After locating these families, municipal governments include them in the Unified Catalog for National Social Programs (pt. CadÚnico), which stores in-depth demographic information about each family. Cash transfers are made through a national bank and are directly transferred to beneficiaries through a Bolsa Família card, which works similarly to a debit card.

Cash transfers are conditional on children's enrollment in schools and on their participation in health care programs, such as immunization. The

¹ *Cinco Vezes Favela*, <https://www.imdb.com/title/tt0055287/>; *City of God*, <https://www.imdb.com/title/tt0317248/>; *Elite Squad*, <https://www.imdb.com/title/tt0861739/>.

² Brazil's Social Communication Secretary. <https://www.gov.br/secom/pt-br/assuntos/noticias/2023/08/bolsa-familia-chega-a-21-14-milhoes-de-familias-em-agosto-241-mil-a-mais-que-em-julho>.

³ The exchange rate is calculated as USD\$1 to R\$4 based on the rate on December 31, 2019. This rate is used throughout the study.

program's reach is considerable mainly because municipalities work closely with beneficiaries through several local reference centers for social assistance (CRAS). Every year, eligibility for renewal of the benefits is assessed by municipal agents working for these centers, who monitor educational results, vaccination rates, and the income level of families. Through the centers, public servants also provide counseling and psychological assistance. Reach and accuracy are two major strengths of Bolsa Família. Nevertheless, the program's national scope makes it rigid to idiosyncratic local features, which often hamper participants' likelihood of progressing out of the program (Wong et al., 2016).

The Banco da Providência Program

In Rio de Janeiro, a non-profit organization named Banco da Providência (BP) acts to complement the Bolsa Família program. Banco da Providência was founded in 1959 by Dom Hélder Câmara, a Brazilian archbishop nominated four times for the Nobel Peace Prize (Câmara, 1968; Bellos, 1999; Condini, 2015). Since its founding, Banco da Providência's mission has been to "reduce extreme poverty in Rio de Janeiro city through professional capability training and income generation for youth, adults, and families that live in poverty situations."⁴ The program is related to the Catholic Church's social movement, and historically, Banco da Providência has operated in more than 60 different favelas in Rio de Janeiro, focusing on those where poverty was particularly intense. According to program reports, their primary goal is to fulfill the United Nations Sustainable Development Goals through thoughtful stakeholder responsiveness at the local level.

Banco da Providência offers a training program that educates, coaches, and supports low-income favela dwellers to improve their psychological morale, develop technical skills, and learn business techniques. The goal is to increase disenfranchised individuals' access to the labor market and to increase income through formal employment or entrepreneurship. The program has two main eligibility criteria. First, the participant's household must have an income level below the poverty line, which is precisely the threshold for qualification for Bolsa Família's cash transfers. Second, the participant must live in a Rio de Janeiro favela where Banco da Providência operates.

The methodology of the Banco da Providência program is inspired by the work of Paulo Freire in the book *Pedagogy of the Oppressed* (1970). According to Freire (1970), the learning process among disenfranchised (i.e., oppressed) people depends heavily on a continuous and balanced dialogue between teacher and student so that the student internalizes their role as the subject of their own story. Moreover, Freire's pedagogy, which uses the Socratic method for developing critical thinking, suggests that the educational approach toward oppressed people should be customized to the reality of participants. This adaptation enables them to understand the world and to enact their liberation (Freire, 1970).

Banco da Providência participants attend two or more weekly training meetings for a total of about nine months. The program has three phases, each

⁴ Banco da Providencia website. <https://www.bancodaprovidencia.org.br/en>.

focusing on a specific capability set. Phase One, entitled Human and Socio-Psychological Capabilities, takes about a month. In this phase, social workers discuss topics such as human and civil rights, domestic violence, communication skills, and communitarian social network dynamics. The objective of Phase One is to improve participants’ morale and psychological preparedness for further training. Phase Two, called Technical Capabilities, takes about seven months. In this phase, instructors teach general and specific technical capabilities in 15 or more different jobs, such as beauty, computing, electrical work, gastronomy, and mechanics. The goal is for each program participant to prepare for a career in one of these domains, according to their preference. Finally, Phase Three, called Managerial and Entrepreneurial Capabilities, takes one month. In this phase, instructors teach managerial skills and strategies to help participants succeed in the labor market, either as employees or entrepreneurs (see Online Appendix 1 for illustrations from the field).⁵ According to a program manager,

We train lots of people to be professionals. However, we develop a professional with a key differential, a training based on socio-psychological characteristics development. We do not train a regular electrician who only knows how to install lamps and stuff. We train an electrician with critical sense, proactivity, solidarity; we use the principles of the UN peace culture. In the end, they go out as a different type of professional. (Banco da Providência General Manager, personal interview)

Table 1 compares the public Bolsa Família and the non-profit Banco da Providência programs.

Table 1. Comparison Between the Public and Non-Profit Programs

	Bolsa Família Public Program	Banco da Providência Non-Profit Program
Activities	Conditional cash transfer on education and health indicators of children.	Socio-psychological support, technical training, and career coaching.
Eligibility	Brazilian families with monthly per capita income below R\$85 (USD\$21.25).	Rio de Janeiro favela families with monthly per capita income below R\$85 (USD\$21.25).
Strengths	Reach: the same program for the whole country, locally implemented by municipal reference centers for social assistance (CRAS). Accuracy: the program targets the ones in most need (hardest to reach).	Customization: the program is explicitly designed to enroll families from the Rio de Janeiro favelas. Flexibility: the program can be adjusted according to the participants’ interests.
Weaknesses	Rigidity and lack of support for participants’ emancipation.	Small-scale with limited geographical reach.

⁵ While Freire’s pedagogy (1970) was developed as a revolutionary literacy method, Banco da Providência’s training program is dedicated to labor market capabilities training and development. Despite this difference in the targeted public, both programs depart from the same epistemological premises and objectives related to the liberation of oppressed people through critical thinking and reasoning.

The Partnership Between Bolsa Família and Banco da Providência

In 2018, Banco da Providência established a public–private partnership with the Municipality of Rio de Janeiro, in which the program was offered to Bolsa Família beneficiaries registered in two municipal reference centers for social assistance (CRAS). The targeted centers are in two of the most violent and vulnerable favelas in the city, Cidade de Deus (City of God) and Pavuna. The partnership’s objective was to run a pilot to determine whether the joint effort would lead individuals to be able to exit the Bolsa Família program. Also, the partnership goals were to foster collaboration between the public Bolsa Família program and the non-profit Banco da Providência program, to stimulate knowledge exchange, and potentially to scale up the partnership to other CRAS agencies in Rio de Janeiro. During a meeting with representatives from Banco da Providência, the two CRAS agencies, and the Rio de Janeiro Social Assistance and Human Rights Secretary, the idea behind the collaboration was stated this way: “the reality is that in the long run, we want to check if this methodology can become public policy” (Banco da Providência Social Assistance Coordinator, meeting at the Rio de Janeiro City Hall).

This study makes use of this partnership to perform a field experiment intended to identify the causal effects of the Banco da Providência program as a complement to the Bolsa Família program. Note that the initial purpose of the experiment, as described in pre-registration, was to evaluate whether a hybrid governance structure—public (BF) and private (BP) organizations in partnership—would lead to superior results compared to a purely public or a purely private governance structure.⁶

TESTING THE TRAINING PROGRAM THROUGH A FIELD EXPERIMENT

Relationship, Data, and Methods

Relationship. The opportunity to gain access to the field site arose through a direct contact from Banco da Providência’s managers. They invited me to assist in evaluating the impact of their training program. Under our agreement, I helped with the design and execution of this evaluation, while in return, the organization authorized the use of data and results for academic research. This collaboration involved no financial transactions, as my work was entirely pro bono. Additionally, the experiment was conducted without any external funding.

Sample. To evaluate the result of the training program, I performed a stratified randomized controlled trial (Bruhn and McKenzie, 2009; Glennerster and Takavarasha, 2013) that compared individuals enrolled in both the Banco da Providência training and Bolsa Família cash transfer programs (hereafter the treatment group) with those enrolled only in the latter cash transfer program

⁶ This research was pre-registered at the American Economic Association Registry for Randomized Controlled Trials (AEA RCT Registry) under protocol number AEARCTR-0002765. All the pre-trial and post-trial information, legal documents signed both by Banco da Providência and the Rio de Janeiro Social Assistance and Human Rights Secretary, dictionary, dataset, and analysis code can be found at <https://www.socialscienceregistry.org/trials/2765>. The study was approved by the University of Toronto Research Ethics Board under protocol number RIS#35741.

(hereafter the control group).⁷ The procedure employed was as follows: Banco da Providência recruited individuals interested in participating in their program with the support of two CRAS agencies, in Cidade de Deus and Pavuna, each of which received Bolsa Família cash transfers. Banco da Providência listed 288 interested individuals, who then participated in a lottery for a spot in the program. Before the lottery, these candidates provided basic information, through which the stratified randomized assignment was executed. The randomization occurred within three strata categories: (i) participants' age, (ii) household income level, and (iii) CRAS location. This process is described further below.

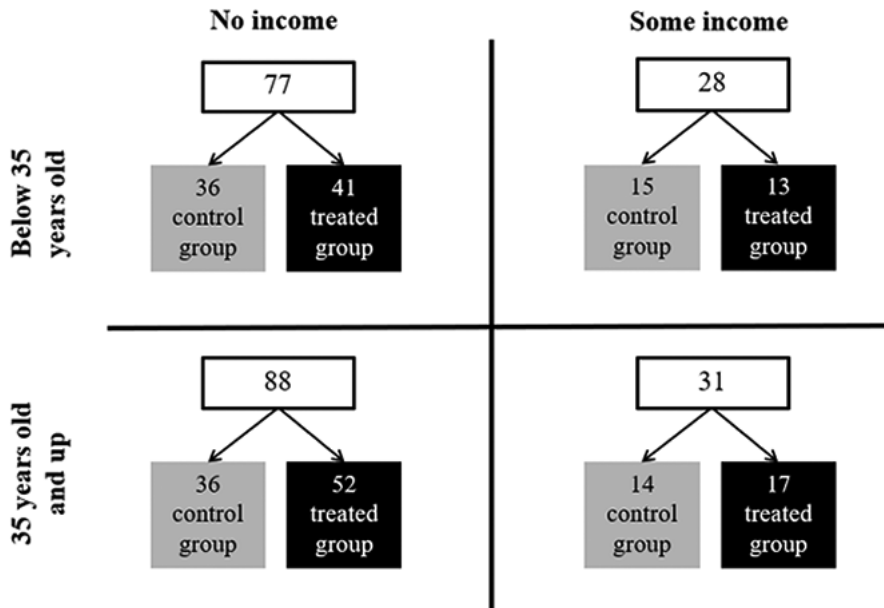
The randomization procedure occurred on March 16, 2018. All methodological steps, as well as the randomization code, were pre-registered before the lottery (see Online Appendix 2 for details). Half of the members of each stratum were allocated to the treatment group and the other half to the control group. This distribution is considered the most appropriate for the research objective, given the limitations of the sample size (Bruhn and McKenzie, 2009).

After the randomization procedure, the staff of Banco da Providência and the CRAS agencies invited all 288 individuals to return to their affiliated CRAS to perform baseline data collection (survey t_0) and to report the results of the lottery. After the survey (t_0), each candidate was told individually whether they had won the lottery and obtained a spot in the program or did not win and would not be able to attend the training program in 2018. Note that 64 individuals did not show up to learn the lottery results and to perform the initial survey. The rates of no-shows were similar across the treatment (35 individuals) and control (29 individuals) groups. As these individuals did not return before learning the result, they were removed from the analysis pool.

Banco da Providência has a quota regarding the number of participants it must enroll per year by location ($N \sim 60$ per location). As a result, a backup waitlist, randomly defined during the stratified randomization procedure, was used to migrate some individuals from the control group to the treatment group. A total of 14 individuals were migrated from the control to the treatment group following the randomly assigned order of the backup waitlist. These individuals were informed and included in the treatment group. As the backup waitlist randomly defined this assignment to the treatment, these participants were considered in the analysis pool as treated individuals.

The training program began on April 2, 2018, after baseline data collection (survey t_0). The initial sample was composed of 101 control and 123 treated individuals. Figure 1 presents the randomization within age by income level stratum. Given the particularities of the setting, a high attrition rate was expected in the program. Indeed, at the end of the nine-month program, after the endline data collection (survey t_1), the experiment had an 8 percent attrition rate, which was balanced over control and treatment. Therefore, the final sample included 207 individuals, of which 93 were in the control group and 114 were in the treatment group. Further analysis of follow-up patterns is presented in the Results section.

⁷ Online Appendix 2 presents additional details about the procedures performed for sampling, data collection, and experimental design.

Figure 1. Stratification Distribution of the Initial Sample (n = 224)*

* For illustration purposes, the diagram does not show the stratification by locality (i.e., Cidade de Deus and Pavuna), only by age and income levels.

Dependent variables. I measured two main economic outcomes. First, the individual's work status was captured as an indicator (0/1) (referred to subsequently as "job type allocation") on one of four dimensions: no income and no job, formal employment, informal employment, or entrepreneurship activity (both formal and informal entrepreneurship). Second, the individual's income or profit measured in Brazilian reais (R\$) per month was assessed on several dimensions: income per capita in the household, total household income, and individual income. Income or profit was directly surveyed with a single question. De Mel and colleagues (2009) showed that asking participants in such programs to self-report income/profit by replying to a single direct question is at least as accurate as asking for revenue and expenses/costs. Furthermore, Banco da Providência social workers (hereafter enumerators) stressed that the direct question enabled participants to understand better what was being asked.

I measured three main socio-psychological characteristics of participants, with scores ranging from 1 to 10. The selection of these three attributes was informed by the paucity of research examining the impact of training programs on these socio-psychological skills, as well as the emphasis on psychological factors during Phase One of the Banco da Providência program. First, following the literature in management and psychology (Sherer and Maddux, 1982; Chen, Gully, and Eden, 2001), I measured an individual's level of self-efficacy. This construct reflected answers to three survey questions on a questionnaire (see the questionnaire in Online Appendix 3). Internal reliability tests performed after final data collection showed a Cronbach's alpha of 0.50 for this construct.

Second, I measured the participant's optimism level. Previous literature studying wealthier individuals justifies use of this measure (Englmaier, 2010; Galasso and Simcoe, 2011). This construct was also measured through answers to three survey questions and had a Cronbach's alpha score of 0.68. Finally, I estimated individuals' awareness/experiences of social stigma related to their place of residence, i.e., favela stigma (Jovchelovitch and Priego-Hernandez, 2013).⁸ This construct was measured from responses to two survey questions, following specialized literature (Pinel, 1999; Kang et al., 2016), with adaptations for this setting. The construct had a Cronbach's alpha score of 0.67.

Training program (treatment). The treatment was captured by a binary variable that takes the value of 1 if the individual received the actual treatment, i.e., enrollment in the Banco da Providência training program while already in the Bolsa Família conditional cash transfer program, and 0 if the individual received only the Bolsa Família conditional cash transfer program. As noted, the characteristics of the setting drove imperfect compliance in our sample, with the treatment variable capturing only the effect of the treatment on compliers. The section below on the estimation model explains the implications in detail.

Fixed effects. Fully controlled models included fixed effects for the randomization strata and training period. For the randomization strata fixed effects, I followed recommendations in the experimental economics literature (Angrist and Pischke, 2008; Bruhn and McKenzie, 2009; Glennerster and Takavarasha, 2013), which suggests that stratification should be performed on variables that could drive heterogeneous effects of the program. To balance the sample and avoid misleading results, the randomized control trial has eight strata categories based on the following variables: (i) age (below 35 years old vs. age 35 and older), (ii) household income (only Bolsa Família as an income source vs. income from this program plus any additional income), and (iii) geographical location of the reference center (Cidade de Deus or Pavuna). In addition, the models included a class/training period dummy to assess potential bias from the training time sessions (morning or afternoon) or possible effects of participants' preferences for particular sessions.

Controls. The models controlled for demographic characteristics, personality traits, and baseline dependent variable scores. The variables in the demographic characteristics set are age, gender ("woman" dummy), race (dummies to reflect the Brazilian Census categorization of the person's identification as of "Black" or "Brown" race), marital status ("single" dummy), number of household members, number of years of schooling, total work experience, total time the individual was unemployed, days worked per week, and hours worked per day. The models included controls for the big five personality traits (John and Srivastava, 1999; Gosling, Rentfrow, and Swann, 2003). Specifically, the study used a reduced, ten-question version of the big five personality traits questionnaire, aiming to control for potential effects that individual personality types

⁸ I also measured poverty, gender, and race stigma. Null results were found for all these stigma types, which serves as a falsification test, reinforcing that spatial stigma is the main driver. This analysis can be found in Online Appendix 4.

would have on participants' job allocation decisions and socio-psychological tendencies (Kerr, Kerr, and Dalton, 2019) (see Online Appendix 3). Finally, the models controlled for dependent variables' baseline values (ANCOVA).⁹ This procedure followed recommendations from experimental economics (Angrist and Pischke, 2008; Glennerster and Takavarasha, 2013).

Estimation model. As the study has imperfect compliance, reflected in an 8 percent attrition rate, the best specification for the analysis is a local average treatment effect (LATE) (Imbens and Angrist, 1994; Angrist and Pischke, 2008), which evaluates and generalizes the causal effect of the treatment for the compliers (the individuals who complied with the *Assignment_i* to the treatment or control group). Note that LATE uses the treatment assignment as an instrumental variable of the actual *Treatment_i*. For an instrumental variable model to hold, three main assumptions are required (Imbens and Angrist, 1994; Angrist and Pischke, 2008): (i) independence, i.e., the treatment assignment must be independent of the outcome and the actual treatment *Assignment_i* $\perp\!\!\!\perp$ *Outcome_{1i}*, *Outcome_{0i}*, *Treatment_{1i}*, *Treatment_{0i}*; (ii) exclusion restriction, i.e., the assignment to the training program affects the outcomes only through the actual treatment *Outcome_i*(*treatment*, 0) = *Outcome_i*(*treatment*, 1) $\equiv$ *Outcome_{di}* for *treatment* = 0, 1; and (iii) monotonicity, i.e., there are no defiers in the program *Treatment_{1i}* – *Treatment_{0i}* $\geq 0 \forall i$.

Given the random assignment to treatment, assumptions (i) and (ii) clearly hold. Concerning assumption (iii), particularly in this setting, two factors support the assumption. First, there is no indication that participants would quit their jobs to enroll in the training program. Historically, dropouts from training arise from the opposite situation, which is that a participant leaves the program to take a job, but there are no instances of such departures in this study. Second, Banco da Providência managers reported that people who do not participate in their program usually do not have alternative training options, and the CRAS staff confirmed this information during interviews. In other words, there are no defiers. Thus, the local average treatment effect assumptions hold. The main analysis model is as follows:

$$Outcome_i = \alpha_i + \hat{\beta} Treatment_i + \sum_{g=1}^8 \delta Strata_{i,g} + \mu Period_i + \gamma' Controls_i + \varepsilon_i$$

$\hat{\beta}$ is the local average treatment effect of the treatment on the compliers for each of the outcomes.¹⁰

$$\hat{\beta} = \frac{E(Outcome_i | Assignment_i = 1) - E(Outcome_i | Assignment_i = 0)}{E(Treatment_i | Assignment_i = 1) - E(Treatment_i | Assignment_i = 0)}$$

$$\hat{\beta} = E(Outcome_{1i} - Outcome_{0i} | Treatment_{1i} \geq Treatment_{0i})$$

⁹ In these types of experiments, analysis of covariance models are preferable to differences-in-differences models with individual fixed effects. Controlling for baseline characteristics increases the power of the test and the accuracy of the estimation. For more information, please check <https://blogs.worldbank.org/impactevaluations/why-difference-difference-estimation-still-so-popular-experimental-analysis>.

¹⁰ Online Appendix 5 presents a sequence of robustness checks that consider potential problems related to multiple hypothesis testing. The majority of the results confirm the main results.

$Outcome_i$ can be several different variables related to economic outcomes and socio-psychological outcomes. $Treatment_i \in \{0,1\}$ is the actual treatment; $Assignment_i \in \{0,1\}$ is the treatment assignment (eligibility for treatment); $\sum_{g=1}^8 \gamma Strata_{i,g}$ is the vector of strata dummies. $Period_i \in \{0,1\}$ is a class training period dummy (1 if morning period, 0 afternoon period). $Controls_i$ is a vector of controls composed of demographic characteristics, personality traits, and baseline (survey t_0) outcome values.

RESULTS

Follow-Up Likelihood

The first set of analyses aims to understand follow-up patterns between groups. In total, 17 people did not follow up for the endline survey (survey t_1), and enumerators gathered qualitative information to obtain reasons for attrition. According to the enumerators, five people were not found (all in the control group), seven people had moved to another neighborhood/city/state (five in treatment and two in the control group), and five individuals were working and could not or did not want to answer the survey at t_1 (four in treatment and one in control).

To make reliable inferences about attrition, the study evaluated whether the treatment had any effect on the likelihood of following up for endline data collection (survey t_1). This test checked whether treated individuals were more or less likely to respond to the endline survey (t_1) after the program ended. Table 2 shows that the treatment had no statistically significant effect on the likelihood of following up in the endline survey (t_1). A joint significance test was performed in the fully controlled model to check whether variables present in the set of controls jointly affected or are correlated with the likelihood of continuing in the sample. Similarly, in this second test, no difference between treatment and control groups is found ($p = 0.959$). In sum, the attrition rate has no relationship with either the treatment or individuals' characteristics.

Following the local average treatment effect estimation procedure, I considered the sample of 207 compliers for which data are available at both baseline (survey t_0) and endline (survey t_1). Table 3 compares the observable

Table 2. Treatment Effect on Participants' Likelihood of Continuing (Following Up)*

	(1) Follow-up	(2) Follow-up
Treatment	0.00604 (0.0358)	-0.00676 (0.0335)
Controls	N	Y
Test of joint significance (F-test)		0.611
p-value		0.959
Observations	224	224
R-squared	0.000	0.198

* Controls include randomization strata fixed effects, training period fixed effects, demographic characteristics, big five personality traits, and outcome values at the baseline (survey t_0). Robust standard errors are in parentheses.

Table 3. Baseline (Survey t_0) Differences Between Treatment and Control Groups*

	Control Mean	Treatment–Control Diff.		Control Mean	Treatment–Control Diff.
Income per capita in t_0 (R\$)	38.04	0.968 (11.66)	Single (dummy)	0.613	0.0625 (0.0672)
Household income in t_0 (R\$)	142.75	–5.849 (42.38)	Education years in t_0	9.301	0.295 (0.394)
Individual income in t_0 (R\$)	29.25	–1.703 (14.59)	Total work experience in t_0 (in months)	50.043	7.738 (10.13)
No income no job in t_0	0.925	0.00509 (0.0365)	Total time is unemployed in t_0 (in months)	22.204	3.655 (5.638)
Formal job in t_0	0.000	0 (0)	Number of household members in t_0	3.409	0.451* (0.213)
Informal job in t_0	0.075	–0.00509 (0.0365)	Q1 Big five personality traits	8.613	0.0889 (0.263)
Entrepreneur in t_0	0.000	0 (0)	Q2 Big five personality traits	4.247	0.261 (0.432)
Self-efficacy in t_0	7.029	0.438* (0.247)	Q3 Big five personality traits	9.323	–0.200 (0.204)
Optimism in t_0	9.251	0.00641 (0.185)	Q4 Big five personality traits	5.817	0.253 (0.480)
Favela stigma in t_0	6.909	0.534 (0.394)	Q5 Big five personality traits	9.323	–0.0682 (0.216)
Days worked per week in t_0	0.441	–0.116 (0.183)	Q6 Big five personality traits	7.161	–0.372 (0.417)
Hours worked per day in t_0	0.817	–0.203 (0.345)	Q7 Big five personality traits	9.032	–0.0673 (0.236)
Age in t_0	35.32	1.300 (1.586)	Q8 Big five personality traits	2.817	0.235 (0.368)
Woman (dummy)	0.860	0.0433 (0.0456)	Q9 Big five personality traits	7.796	–0.0589 (0.370)
Black (dummy)	0.376	0.0535 (0.0687)	Q10 Big five personality traits	3.914	–0.0894 (0.438)
Brown (dummy)	0.462	–0.0413 (0.0697)			
Test of joint significance (F-test)		0.941			
p-value		0.555			
Observations	93	114			

+ $p < .10$; * $p < .05$.

* Robust standard errors are in parentheses.

characteristics of these individuals at the baseline. If the randomization process was successful, the two samples, i.e., treatment and control, should be statistically similar. This is confirmed, as the two groups are statistically identical based on a joint significance test ($p = 0.555$).

Table 3 lists the main characteristics of the final sample. For both control and treated individuals, the average monthly income per capita is R\$39 (USD\$9.75); 93 percent of individuals have neither income nor a job; approximately 88 percent are women; 84 percent are persons of color; and 65 percent are single. These descriptive statistics suggest the high intersectionality of participants in terms of gender, race, and socioeconomic condition (Crenshaw, 2017). The sample characteristics are also representative of Brazilian favela households, which are usually composed of single, unemployed mothers of color (Perlman, 2010; Jovchelovitch and Priego-Hernandez, 2013; Meirelles and Athayde, 2016). Taken together, these features suggest the potential generalizability of the results.

Table 4. Treatment Effect on Job Allocation of Participants*

DVs =	(1) No income no job in t_1	(2) No income no job in t_1	(3) Informal job in t_1	(4) Informal job in t_1	(5) Formal job in t_1	(6) Formal job in t_1	(7) Entrepreneur in t_1	(8) Entrepreneur in t_1
Treatment	-0.112 (0.0698)	-0.146+ (0.0779)	-0.0526 (0.0648)	-0.0230 (0.0730)	0.0416 (0.0465)	0.0509 (0.0533)	0.123** (0.0309)	0.118** (0.0339)
Constant	0.559** (0.0517)	1.328* (0.602)	0.333** (0.0491)	0.171 (0.511)	0.108** (0.0323)	-0.155 (0.445)	0	-0.344 (0.342)
Controls	N	Y	N	Y	N	Y	N	Y
DV average (in %)	49.76%		30.44%		13.04%		6.76%	
Observations	207	207	207	207	207	207	207	207
R-squared	0.012	0.191	0.003	0.148	0.004	0.118	0.059	0.305

+ $p < .10$; * $p < .05$; ** $p < .01$.

* Controls include randomization strata fixed effects, training period fixed effects, demographic characteristics, big five personality traits, and outcome values at the baseline (survey t_0). Robust standard errors are in parentheses.

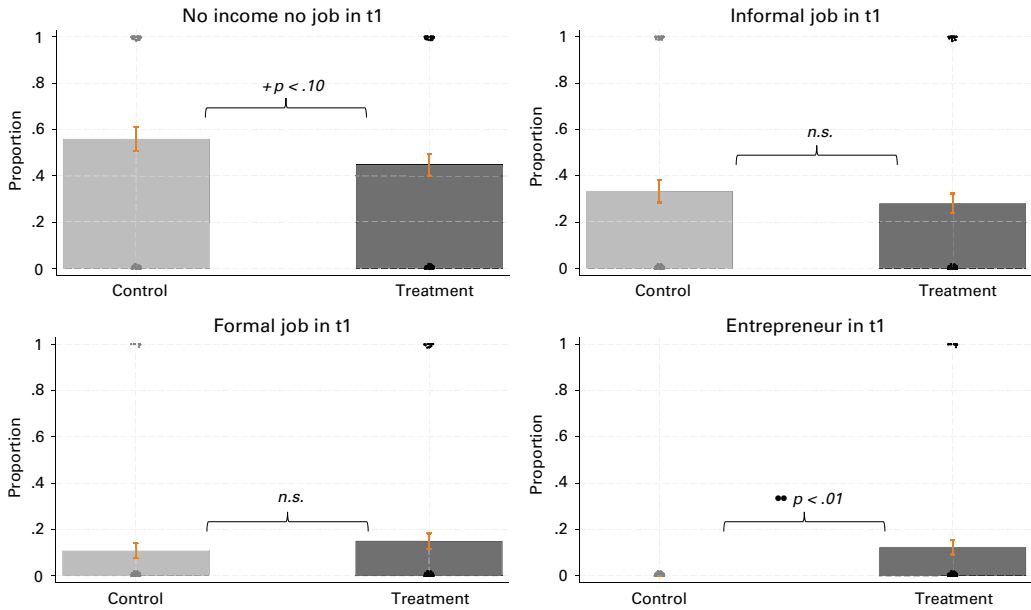
Economic Outcomes

The next step is to understand whether the treatment had an effect on participants' job allocation and, if so, on which types of jobs. Table 4 shows that the treatment had a marginally significant effect ($p < 0.1$) on diminishing the "no income, no job" condition. When compared to the control group, treated individuals were 14.6 percentage points more likely to have income and a job after the program ($p < 0.1$). Analyzing the job allocation types reveals that neither informal nor formal employment shows any statistical effect of the treatment.¹¹

The results show a strong significant effect of the treatment on entrepreneurship development, 11.8 percentage points ($p < 0.01$). In other words, the main path through which the treatment reduced the "no income, no job" condition was through small-scale entrepreneurship. Note that both formal and informal forms of entrepreneurship were considered. Figure 2 presents these results. This boost in entrepreneurship may occur as a constrained career choice (Rider et al., 2019; Dencker et al., 2021), which is a fallback option to more-rewarding and less-risky options, such as formal employment (Hwang, 2022; Hwang and Phillips, 2023).

The absence of entrepreneurs in the control group at the study's endline may seem unexpected, particularly given the prevalence of necessity entrepreneurship in countries like Brazil. There are two main reasons for this result. First, without exposure to the training program, control group individuals often overlooked entrepreneurship as a viable career option. Second, the stark lack of resources among control individuals played a crucial role. In contrast, during their training, treated individuals might have had access to essential resources such as toolkits (e.g., appliances, machines, and tools) and production inputs (e.g., cooking ingredients and sewing materials), providing a foundational platform for initiating their entrepreneurial ventures.

¹¹ Notice that despite being statistically equivalent, individuals in the treated group have informal and formal jobs with higher pay. Data show that the income per capita in t_1 in informal jobs equals R\$239.65 (controls) and R\$300.03 (treated), and in formal jobs it equals R\$405.21 (controls) and R\$554.32 (treated). In other words, it seems that the treated group usually had better jobs both informally and formally.

Figure 2. Job Type Allocation After the Treatment*

+ $p < .10$; * $p < .05$, ** $p < .01$; non-significant (n.s.).

* The y-axes of the graphs are on similar scales. Since all variables are dummy variables, scattered points fall on the extremes. The plots use raw data, and p -values consider the fully controlled models. Data points are slightly jittered to enable an accurate interpretation of the data distribution.

Table 5 shows that the treatment had a significant effect on income generation among participants. On average, the income per capita increased by R\$106.30 (USD\$26.5) per month ($p < 0.01$). This is a 0.46 standard deviation increase in income per capita among the treated relative to control participants. On average, this change represents 2.7 times the pre-treatment monthly income per capita of the treated group.¹² The total household income increased by R\$322.50 (USD\$80.75) per month ($p < 0.01$). This is a 0.43 standard deviation increase in household income among the treated relative to control participants. On average, this change represents 2.4 times the pre-treatment monthly household income of the treated.¹³ Finally, the individual income (i.e., of the surveyed participants) increased by R\$154.00 (USD\$38.5) per month ($p < 0.05$). This is a 0.32 standard deviation increase in the individual income among the treated relative to control participants. On average, this change represents 5.6 times the pre-treatment monthly individual income of the treated.¹⁴

¹² Tables 3 and Table 5, such that $EffectSize^T = [\hat{\beta}^T / AvgDV_0^T] = (R\$106.30) / (R\$38.04 + R\$0.968) = 2.73 \equiv 273\%$.

¹³ Tables 3 and Table 5, such that $EffectSize^T = [\hat{\beta}^T / AvgDV_0^T] = (R\$ 322.50) / (R\$142.80 - R\$5.849) = 2.35 \equiv 235\%$.

¹⁴ Tables 3 and Table 5, such that $EffectSize^T = [\hat{\beta}^T / AvgDV_0^T] = (R\$154.00) / (R\$29.25 - R\$1.703) = 5.59 \equiv 559\%$.

Table 5. Treatment Effect on Income Generation of Participants*

DVs =	(1) Income per capita in t1 (R\$)	(2) Income per capita in t1 (R\$)	(3) Household income in t1 (R\$)	(4) Household income in t1 (R\$)	(5) Individual income in t1 (R\$)	(6) Individual income in t1 (R\$)
Treatment	98.65** (30.71)	106.3** (34.25)	351.2** (97.59)	322.5** (104.1)	133.2● (65.92)	154.0● (73.00)
Constant	159.8** (18.60)	57.58 (256.5)	565.7** (51.96)	165.3 (970.8)	304.0** (45.49)	-326.8 (512.8)
Controls	N	Y	N	Y	N	Y
DV average (in R\$)	R\$214.08		R\$759.14		R\$377.33	
Observations	207	207	207	207	207	207
R-squared	0.045	0.238	0.054	0.226	0.019	0.239

● $p < .05$; ** $p < .01$.

* Controls include randomization strata fixed effects, training period fixed effects, demographic characteristics, big five personality traits, and outcome values at the baseline (survey t_0). The calculation of the treatment effect sizes among treated follows the formula $EffectSize^T = [\hat{\beta}^T / AvgDV_0^T]$. Robust standard errors are in parentheses.

It is essential to be cautious about using individual income as the primary income measure. A substantial number of individuals work only occasionally as opportunities emerge or work in small family businesses. Thus, it is difficult for them to estimate their individual income precisely. Several enumerators reinforced this point based on the data collection surveys. Therefore, a total household income or an income per capita measure provides a better understanding of the monetary position of the entire household unit. This is also the reference unit that both the Banco da Providência and Bolsa Família programs employ. Figure 3 presents the income comparison between the treated and control groups.

Socio-Psychological Outcomes

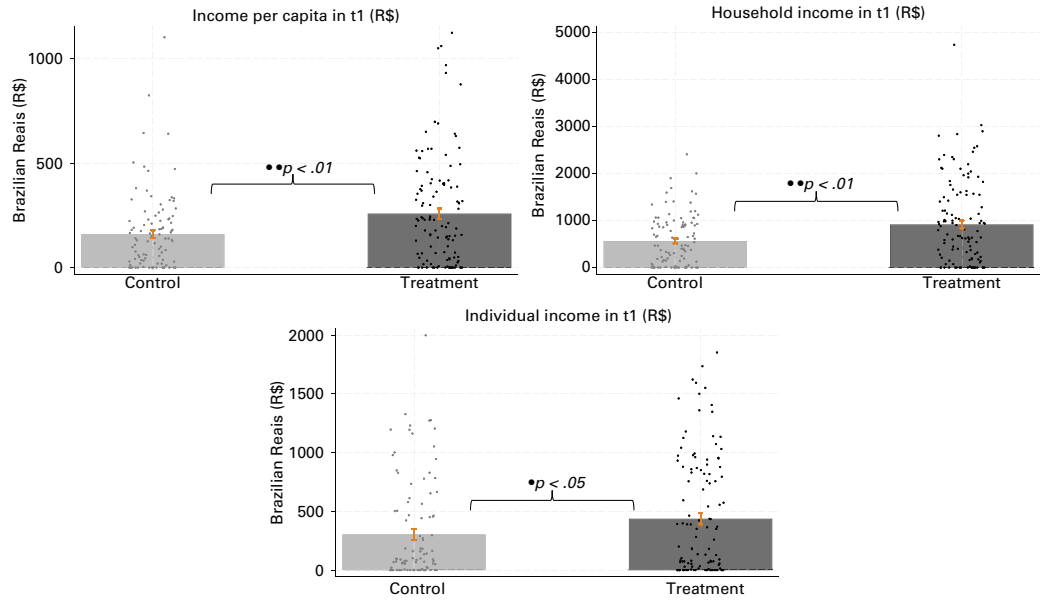
Table 6 shows that the treatment has a significant effect on the socio-psychological outcomes of participants. On average, participants increased their self-efficacy by 0.524 points ($p < 0.05$) and their optimism level by 0.409 points ($p < 0.05$). These represent a 0.32 standard deviation increase in self-efficacy and a 0.29 standard deviation increase in optimism among the treated relative to control participants. On average, these changes represent a 7 percent increase in self-efficacy and a 4.4 percent increase in optimism among the treated individuals.¹⁵

However, the treatment group reported greater experience of stigma based on their favela residence (0.742 points) than did those in the control group ($p < 0.05$). This is a 0.27 standard deviation increase in favela stigma among the treated relative to control participants. On average, this change represents a 10 percent increase in experiences of favela stigma among the treated group, which is approximately equivalent to moving one decile up in the stigma distribution.¹⁶ Figure 4 illustrates the socio-psychological comparison between the treated and control groups.

¹⁵ Tables 3 and Table 6, such that $EffectSize^T = [\hat{\beta}^T / AvgDV_0^T] = (0.524) / (7.029 + 0.438) = 0.070 \equiv 7.0\%$. And: $EffectSize^T = [\hat{\beta}^T / AvgDV_0^T] = (0.409) / (9.251 + 0.00641) = 0.044 \equiv 4.4\%$.

¹⁶ Tables 3 and Table 6, such that $EffectSize^T = [\hat{\beta}^T / AvgDV_0^T] = (0.742) / (6.909 + 0.534) = 0.100 \equiv 10.0\%$.

Figure 3. Monthly Income After the Treatment*



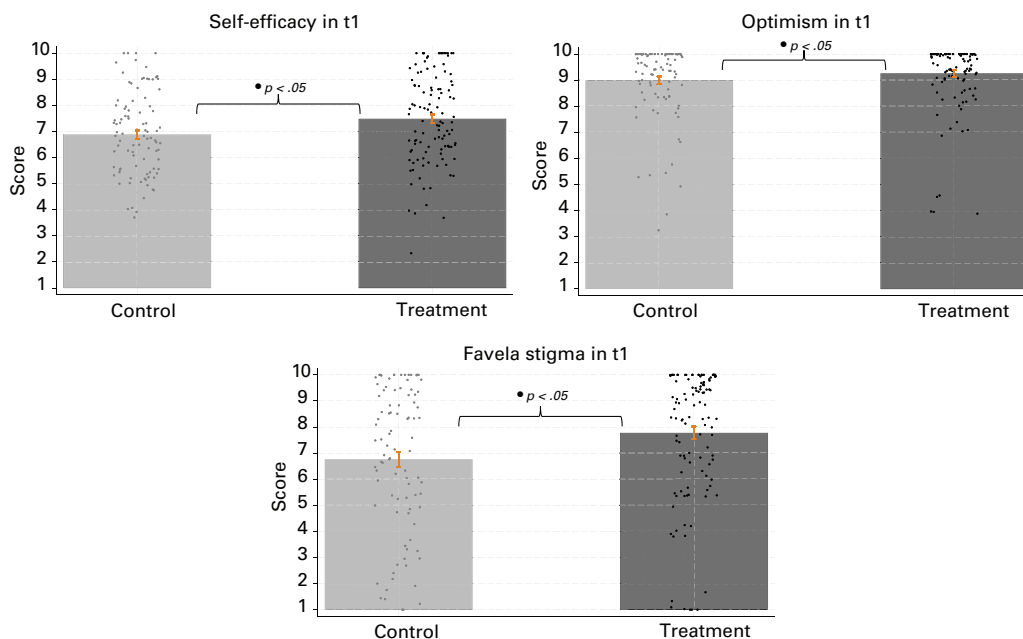
* $p < .05$; ** $p < .01$.
* The y-axes of the graphs are on different scales. The plots use raw data, and p -values consider the fully controlled models. Data points are intensely jittered to enable better visualization of the data distribution.

Table 6. Treatment Effect on Socio-Psychological Aspects of Participants*

DVs =	(1) Self-efficacy in t1	(2) Self-efficacy in t1	(3) Optimism in t1	(4) Optimism in t1	(5) Favela stigma in t1	(6) Favela stigma in t1
Treatment	0.604** (0.226)	0.524* (0.217)	0.264 (0.197)	0.409* (0.190)	1.018** (0.381)	0.742* (0.369)
Constant	6.882** (0.160)	2.619 (2.011)	8.993** (0.151)	2.599 (1.706)	6.758** (0.296)	5.389* (2.711)
Controls	N	Y	N	Y	N	Y
DV average (in score)	7.214		9.138		7.319	
Observations	207	207	207	207	207	207
R-squared	0.033	0.285	0.009	0.288	0.034	0.326

* $p < .05$; ** $p < .01$.
* Controls include randomization strata fixed effects, training period fixed effects, demographic characteristics, big five personality traits, and outcome values at the baseline (survey t_0). Social-psychological variables are measured on a scale ranging from 1 to 10 points (see Online Appendix 3). The calculation of the treatment effect sizes among treated follows the formula $EffectSize^T = [\beta^T / AvgDV_0^T]$. Robust standard errors are in parentheses.

It is complex to describe the social significance of the favela stigma result as this would represent an attempt to measure the extent to which an emotion affects an individual's social-psychological state. To provide a refined

Figure 4. Socio-Psychological Outcomes After the Treatment*

* $p < .05$.

* The y-axes of the graphs are on similar scales. The plots use raw data, and p -values consider the fully controlled models. Data points are intensely jittered to enable a better visualization of the data distribution.

approximation, consider that before a certain educational program, you were aware of ten individuals known to harbor prejudices against favela residents. After the program ends, you encounter an additional individual who displays bias against you due to your residence in a favela. Suppose this individual is a potential employer assessing you during a recruitment process or a supplier from whom you intend to procure products for your newly launched business venture. What emotional response would this encounter elicit in you? What would be the social significance of it?

DO INCOME GAINS AND FAVELA STIGMA GO HAND IN HAND? UNVEILING THE ALLEGORY OF THE FAVELA

The results of this study's analysis of favela stigma run opposite to what was predicted in pre-registration, which itself can be considered a contribution (it was predicted that the treatment would lead to reduction in the favela stigma). Two competing explanations exist for this unexpected result. The first explanation for the increase in individuals' experience of stigma is that the training program was biased or discriminatory against the participants. If so, the treatment itself would have generated negative emotions and experiences of stigmatization among participants. An alternative explanation for this increased experience of stigma is that after participating in the program, many individuals searched for employment outside the favela. Through heightened interactions with peers outside the favela, these individuals faced prejudice more intensely.

To test for these alternative competing explanations and to comprehensively understand the mechanisms behind the surprising result regarding favela stigma, I departed from the pre-registered study description and performed a sequence of additional tests in order to generate a theoretical contribution through abduction. Thus, I conducted the following non-preregistered additional empirical exercises: (i) computational text analysis of participants' feedback during the training program, (ii) a set of quantile regression models evaluating the correlation between the favela stigma scores and income gains at different moments of the income distribution, and (iii) triangulation of the quantitative results with qualitative evidence from field visits and interviews with participants. This section also unveils the "allegory of the favela," which informs the study's theoretical implications.

Computational Text Analysis

Computational text analysis of participants' feedback during the training program reveals non-significant correlations between participants' positive or negative emotions during the program (at the end of Phase One) and their experiences of favela stigma after the program ended (survey t_1) (see Online Appendix 6 for more details). In other words, these results suggest that the training program did not trigger the stigmatization experiences among participants. However, this evidence does not corroborate the above-noted second explanation as most adequate.

Quantile Regressions

To properly evaluate which of the two explanations is the best conjecture for the increased favela stigma at the program's end, I used a quantile regression model (Koenker and Bassett, 1978; Abrevaya, 2001; Koenker and Hallock, 2001; Firpo, 2007). This estimation examined how the favela stigma's relationship with income changes conditional on which quantile of the income distribution individuals are in at the endline (survey t_1).¹⁷ This analysis helps to shed light on the discrimination mechanism given that individuals with higher socioeconomic mobility (i.e., in higher quantiles of the income distribution) tend to interact more frequently with relatively higher-status individuals (employers and suppliers) beyond the favela boundaries.

This estimation approach has several advantages. A quantile regression model enables assessment of the strength in the correlation between these two variables on various segments of the income distribution, i.e., when comparing lower and higher levels of socioeconomic mobility. Unlike an OLS analysis, this approach offers more-comprehensive understanding of the relationship between two variables because it does not presume a constant effect throughout all parts of the dependent variable distribution. Therefore, it properly allows evaluation of whether income and stigma go hand in hand, according to the strength of their association at different income levels.

Moreover, quantile regressions are an attractive alternative estimation when OLS models are not the best linear unbiased estimator, i.e., when OLS models

¹⁷ Note that the stigmatization process does not vary according to participation in the training program, so it is expected to affect both the treated and control groups comparably. Thus, these analyses included the entire sample of participants.

suffer from non-normality of residuals and heteroscedasticity. This is precisely the case in this study (see Online Appendix 7 for more details). Following Koenker and Bassett (1978: 38), the quantile regression minimizes the following equation with respect to β :

$$\min_{\beta \in \mathbb{R}^K} \left[\sum_{i \in \{Income_i \geq Favela_i' \beta + X_i' \gamma\}}^N q |Income_i - Favela_i' \beta - X_i' \gamma| + \sum_{i \in \{Income_i < Favela_i' \beta + X_i' \gamma\}}^N (1 - q) |Income_i - Favela_i' \beta - X_i' \gamma| \right]$$

where $Income_i$ is the dependent variable of the model measuring the income per capita of the individual i at the endline (survey t_1); $Favela_i$ is the explanatory variable favela stigma of the individual i at the endline (survey t_1); X_i is a vector of control variables including job allocation at the endline (t_1), income level at the baseline (t_0), and demographic characteristics at the baseline (t_0); and q is the estimated quantile. The main models consider each income decile ($q10 \leq q \leq q90$).

The models were estimated both with the favela stigma variable and with its factors isolated, i.e., the two statements composing the favela stigma construct. Therefore, in Table 7 and Figure 5, Panel (A) evaluates the relationship between favela stigma in t_1 and income per capita in t_1 ; Panel (B) analyzes this relationship using favela stigma's first factor, which states, "I often suffer from prejudice because I live in a favela" in t_1 ; and Panel (C) considers this relationship using favela stigma's second factor, which states, "I believe I have a harder time getting a job because I live in a favela" in t_1 . These individual factors help to determine whether income gains and favela stigma go hand in hand because of participants' experiences of prejudice (Panel B) or participants' difficulties in successfully finding work (Panel C).

Table 7, Panel (A) demonstrates that the favela stigma in t_1 significantly differs among lower- versus higher-income deciles. From the tenth income decile ($q10$) to the median income decile ($q50$), the favela stigma is approximately zero or slightly negative and non-significant. By the sixth income decile ($q60$), the favela stigma coefficient is 3.22 but still non-significant. At the seventh income decile ($q70$), the favela stigma coefficient is 8.87 ($p < 0.1$), and by the eighth income decile ($q80$) the coefficient is 13.41 ($p < 0.1$). Finally, the highest income decile ($q90$) has a coefficient of 21.63 ($p < 0.05$), which is approximately 13 times higher than Panel (A)'s OLS coefficient. Figure 5, Panel (A) graphically illustrates these results.

The results become even more striking when the favela stigma construct is decomposed along its factors. Comparing Table 7, Panels (B) and (C), we observe that both indicators increase in magnitude as we move to higher-income deciles. However, only Panel (B), which measures experiences of prejudice more directly, reveals significant increases with higher magnitudes. Specifically, Panel (B) shows that at the seventh income decile ($q70$) the coefficient equals 7.38 ($p < 0.1$), at the eighth income decile ($q80$) the coefficient is 10.66 ($p < 0.1$), and at the highest income decile ($q90$) the coefficient equals 23.54 ($p < 0.05$), which is approximately six times higher than

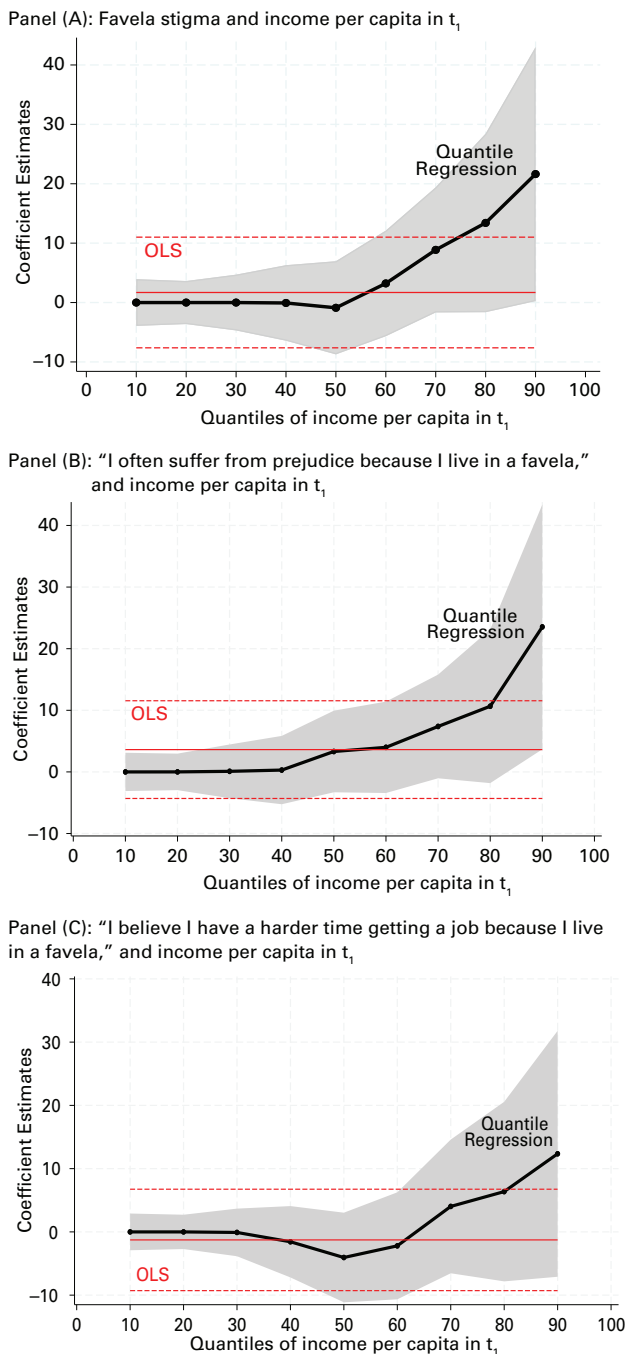
Table 7. OLS and Quantile Regression Between Favela Stigma in t_1 and Income Per Capita in t_1 (R\$)*

DV =	(0)	(1)	(2)	(3)	Income Per Capita in t ₁ (R\$)					(7)	(8)	(9)
	OLS	q10	q20	q30	q40	q50	q60	q70	q80	q90		
Panel (A): Favela stigma in t ₁	1.693 (4.641)	0 (1.995)	-0.00122 (1.837)	-0.00181 (2.381)	-0.0690 (3.223)	-0.884 (3.975)	3.223 (4.508)	8.872 ⁺ (5.342)	13.41 ⁺ (7.621)	21.63 [•] (10.84)		
Constant	175.6 ⁺ (94.64)	0 (51.16)	2.387 (54.05)	18.72 (64.44)	42.33 (78.39)	101.0 (90.32)	117.5 (99.42)	138.8 (118.3)	148.8 (169.3)	234.4 (247.5)		
Panel (B): "I often suffer from prejudice because I live in a favela" in t ₁	3.622 (4.087)	0 (1.575)	0.00405 (1.502)	0.0928 (2.210)	0.298 (2.810)	3.314 (3.356)	3.991 (3.759)	7.378 ⁺ (4.265)	10.66 ⁺ (6.324)	23.54 [•] (10.07)		
Constant	166.8 ⁺ (94.24)	0 (50.03)	2.030 (59.28)	16.73 (73.13)	35.86 (86.68)	76.68 (95.04)	120.8 (104.7)	157.3 (120.7)	204.5 (165.7)	162.2 (255.0)		
Panel (C): "I believe I have a harder time getting a job because I live in a favela" in t ₁	-1.267 (3.904)	0 (1.471)	0 (1.375)	-0.0848 (1.908)	-1.560 (2.858)	-4.049 (3.590)	-2.204 (4.287)	4.025 (5.356)	6.371 (7.195)	12.35 (9.855)		
Constant	191.8 [•] (95.99)	0 (53.10)	0 (56.84)	19.84 (74.69)	61.00 (83.48)	117.1 (93.40)	143.4 (107.0)	134.2 (128.6)	169.6 (186.3)	245.4 (245.4)		
Controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y		
Observations	207	207	207	207	207	207	207	207	207	207		
Bootstrap replications	N/A	500	500	500	500	500	500	500	500	500		
Descriptive statistics												
DV average by quantile (in R\$)	R\$214	R\$ 0	R\$ 0	R\$45	R\$97	R\$160	R\$216	R\$297	R\$422	R\$733		
Job allocation by quantile in t ₁												
No income, no job (in %)	49.8%	100.0%	100.0%	62.5%	65.2%	26.1%	37.5%	31.8%	8.0%	4.8%		
Informal job (in %)	30.4%	0.0%	0.0%	37.5%	26.1%	56.5%	54.2%	36.4%	36.0%	38.1%		
Formal job (in %)	13.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	22.8%	44.0%	42.9%		
Entrepreneur (in %)	6.8%	0.0%	0.0%	0.0%	8.7%	17.4%	0.0%	9.1%	12.0%	14.3%		
Number of individuals in each quantile	207	27	26	16	23	23	24	22	25	21		

+ $p < .10$; * $p < .05$.

* Controls include individual-level characteristics, namely job allocation at the endline (survey t_1), income level at the baseline (survey t_0), and demographic characteristics at the baseline (survey t_0). Socio-psychological and personality trait baseline controls were not included to avoid bad-control issues, given their strong intertemporal correlation. Models were estimated using 500 bootstrap replications. The first and second quantiles represent all individuals with zero income per capita in t_1 (53 in total). Panel (C) had 206 respondents at the endline (survey t_1). In the lower part of the table, darker shades indicate higher values. Robust standard errors are in parentheses.

Figure 5. OLS and Quantile Regression Between Favela Stigma in t_1 and Income Per Capita in t_1 (R\$)*



* Point estimates (solid lines) and 95% C.I. (shaded area and dashed lines). OLS is represented with thin lines, quantile regression is represented with thick lines and shaded areas. All results are controlled by individual-level characteristics, namely job allocation at the endline (survey t_1), income level at the baseline (survey t_0), and demographic characteristics at the baseline (survey t_0). Given their strong intertemporal correlation, the socio-psychological and personality trait baseline controls were not included to avoid bad-control issues. Panel (C) had 206 respondents at the endline (survey t_1). Models estimated using 500 bootstrap replications.

Panel (B)'s OLS coefficient. Figure 5, Panels (B) and (C) graphically illustrate these results.

Note that in higher-income deciles, individuals perform jobs that usually involve more interactions with peers from outside favelas. At the bottom of Table 7, we see that at the seventh decile ($q70$), 31.9 percent of individuals are either formally employed (22.8 percent) or developing small enterprises (9.1 percent). At the eighth income decile ($q80$) this joint percentage increases to 56.0 percent, and at the highest income decile ($q90$) this joint proportion equals 57.2 percent, the highest amount among all income deciles. In other words, the higher the socioeconomic mobility (higher quantiles), the more likely individuals are to interact with external peers of relatively higher status.¹⁸ These results support the second explanation, suggesting that individuals going outside favelas for employment opportunities or searching for suppliers have higher income levels, and as income levels increase, participants' experiences of discrimination also increase.

Qualitative Evidence

Qualitative data were collected in July 2019, seven months after the program ended, through 14 semi-structured recorded interviews with participants and community leaders, as well as meetings and roundtables with enumerators, CRAS employees, Rio de Janeiro Social Assistance and Human Rights social workers and administration, and Banco da Providência managers. In total, the data amount to 764 minutes of audio recordings. The informants were selected by convenience (Strauss and Corbin, 1990). These interviews, meetings, and roundtables were complemented by two field visits, one in each favela in the sample, during which I visited some of the local enterprises developed by participants.

Data triangulation between the previous quantitative results and qualitative information validates the second explanation regarding the multifaceted nature of socioeconomic mobility. Informants provided several illustrative cases and descriptions supporting this explanation. One enumerator mentioned the case of a participant in the final phases of a formal employment recruitment process, at which point her application was declined after she notified the potential employer of her address inside Cidade de Deus. Similarly, during one of the field visits, an entrepreneur from Pavuna reported having difficulties purchasing and receiving supplies when describing the location of her store inside the favela, a phenomenon that I documented in earlier research (Pongeluppe, 2022). Several informants described the process of searching for employment and suppliers outside favelas as "getting out of the ghetto." During an interview, when asked directly about possible prejudice toward favela dwellers, another entrepreneur stated,

Almost nobody wants to invest in a community [favela], nobody looks at a community [favela] and says "wow, good things will arise from there." (Program participant and entrepreneur from Cidade de Deus, personal interview)

¹⁸ Interestingly, the favela stigma is the only stigma that significantly soared with income gains. Online Appendix 8, Panel (B) shows the results of quantile regressions on alternative types of stigma, namely poverty, gender, and race. None of these alternative stigmas significantly change for higher-income quantiles.

Finally, a community leader from Cidade de Deus gave her interpretation of why the prejudice against favela populations is so fierce:

The violence hinders [favela residents] a lot, so when people say they feel discriminated against because they live in the favela, it is precisely for this reason. When you say: "I live in a favela," people already look down on you. Because automatically, those living in a favela are either drug dealers or gang members. To people from outside the favela, there is no possibility that you are just a resident. However, the truth is that we coexist but do not collude [with the gangs]. We can't do anything to change this situation except to resist. (Community leader from Cidade de Deus, personal interview)

The Allegory of the Favela

An "allegory" is frequently described as a story within a story. Inspired by Plato's allegory of the cave in *The Republic* (Bloom, 1968), the allegory of the favela provides a symbolic narrative of the manifest and latent experiences of favela dwellers as they navigate societal perceptions in their quest for socio-economic mobility. Initially, individuals residing within these stigmatized neighborhoods lacked external social references, leaving them largely unaware of the extent of their devalued societal status and the depth of their stigmatization. Their interactions, primarily confined to the favela, established a socio-psychological comfort zone while restricting opportunities for broader educational and socioeconomic advancement.

Post-training, these individuals emerged equipped with enhanced skills, ready to enter the labor force. However, due to their devalued social status, their endeavors often exposed them to the harsh realities of societal prejudice, juxtaposing the gains of economic mobility against the challenges of discrimination. This bittersweet process forced many individuals to pivot toward entrepreneurship within their communities. This shift can foster their economic progress and position them as agents of change, potentially liberating others within their community from economic disenfranchisement while partially safeguarding them from external discriminatory systems.

The allegory of the favela shows that socioeconomic mobility can change how much individuals earn but not where they come from or how society reacts to them. Similar to the ancient allegory of Plato (Bloom, 1968), the process experienced when getting out of the favela is generalizable to other contexts and situations. Examples include immigrants and ethnic groups who are economically integrated but discriminated against due to their country of origin or religion (Marinoni, 2023), first-generation students and first-generation white-collar workers who develop their academic and labor skills but are underappreciated due to their working-class backgrounds and lack of "pedigree" (Rivera, 2015: 3), and gender and racial minorities who advance in their professional careers but are deemed unfit and unable (Smith, 2005; Kanter, 2008). The allegory of the favela symbolically illustrates and generalizes the multifaceted process of socioeconomic mobility experienced by individuals of devalued social status, who, despite prospering economically, realize through their inclusion in social systems that those systems are biased and discriminatory against them (see Online Appendix 9 for more details).

IMPLICATIONS FOR THEORY AND PRACTICE

The findings of this study have several implications for the literature on labor market bias and discrimination and on social comparison, inclusion, and belonging.

Labor Market Bias and Discrimination

In the same vein as Hwang (2022) and Hwang and Phillips (2023), this study demonstrates that entrepreneurship can be a potential career path for individuals facing labor market discrimination. Although the individuals in this study did not have criminal records, their place of residence in the favelas affected their likelihood of entering formal labor markets in a way similar to the experiences of formerly incarcerated individuals (Hwang, 2022; Hwang and Phillips, 2023). In other words, discrimination from potential employers, who classified favela dwellers as potentially dangerous and associated with criminal activities, hurt these individuals' chances of entering the labor market as formal employees, whose jobs usually represent more-stable, less-risky, and higher-paid careers. The results reinforce the idea that necessity entrepreneurship (Dencker et al., 2021) can be an appealing alternative career path given its effects on socioeconomic mobility. Disenfranchised individuals might considerably increase their income by engaging in these small-scale enterprises. Also, these endeavors might contribute to the economic development of stigmatized neighborhoods (Ropert and Di Masso, 2021; Born, 2023), fostering market activities in communities that have traditionally been underserved by conventional firms (Pongeluppe, 2022).

The findings in this study are also theoretically significant for the literature by highlighting the concept of location-based discrimination, an often-understudied type of discrimination. Studies in labor market discrimination traditionally have focused on individuals' observable characteristics, such as gender and race, when evaluating bias and discrimination in hiring and promotion practices (Goldin and Rouse, 2000; Kang et al., 2016; He, Kang, and Lacetera, 2021; Rider et al., 2023). In line with studies that consider non-observable characteristics such as social class (Rivera, 2015) and ethnic background (Marinoni, 2023) as potential sources of labor market discrimination, this study enlarges this set of possibilities by considering another non-observable characteristic: place of residence. To my knowledge, this is the first study to experimentally evaluate the effect of individuals' place of residence within segregated and stigmatized neighborhoods (Massey and Denton, 1988; Charles, 2003; Ropert and Di Masso, 2021; Born, 2023) on their likelihood of entering the labor market.

This study's findings are also theoretically significant for the literature on training programs dedicated to disenfranchised people. Training programs have been extensively deemed one of the best tools to promote socioeconomic mobility. By intervening on the supply side of the market, training programs provide individuals with knowledge and skills that enable them to improve their economic conditions (Attanasio, Kugler, and Meghir, 2011; de Mel, McKenzie, and Woodruff, 2012; McKenzie and Woodruff, 2014; Bulte, Lensink, and Vu, 2017; Carlson and Hager, 2021). This study confirms the adequacy of training programs to promote economic gains for disenfranchised individuals, but these interventions do not affect the demand side of the market, i.e., how societal

systems will receive and include the newly trained individuals. The study shows that socio-psychological outcomes might be mixed. Therefore, interventions focused on changing the supply-side behavior might more effectively promote liberations (Freire, 1970) if such interventions are informed by more-extensive consideration of demand-side perceptions and responses.

Social Comparison, Inclusion, and Belonging

The study also has interesting implications for the social comparison literature by reinforcing well-known mechanisms described in this literature. Individuals change their social comparison set by interacting with higher-status peers (Merton, 1968; Frank, 1985). In changing their reference group for comparison, individuals (and organizations) are more aware of their relative social position and status and update their evaluation of themselves based on their peers' perceptions (Tan and Rider, 2023). This literature has conventionally focused on high-end settings such as Nobel laureates and immediately close peers (Merton, 1968), law firms and legal services (Phillips and Zuckerman, 2001; Rider and Tan, 2015; Tan and Rider, 2017), venture capital funds and financial advice (Phillips and Zuckerman, 2001; Rider, 2009), and private colleges and universities (Askin and Bothner, 2016). Thus, this article also contributes by presenting evidence that similar mechanisms operate in a low-end setting. As Freire (1970: 63) suggested, "Self-depreciation is another characteristic of the oppressed, which derives from their internalization of the opinion the oppressors hold of them."

This study's findings are of interest to the literature on inclusion and feelings of belonging. Similarly to Turco (2010), this study shows that discriminatory practices are subject to local cultural context and contingencies. As described in the quote above by the Cidade de Deus community leader, violence associated with favelas—which is stereotypically pictured even in movies—hinders favela populations' opportunities for development. This process happens in two ways. First, this association neglects favela dwellers' access to educational, cultural, and financial support systems. Second, among potential employers and suppliers outside favelas, it fosters discriminatory views about these populations and their communities. This cultural context of prejudice against favelas limits people's entry into the labor market, causes socio-psychological distress due to their stigmatization, and may foster "impostor syndrome" feelings among them (Tewfik, 2022: 1011). As Kanter (2008: 9) described, "The loss of a job, or the right to pursue a profession, or the opportunity to continue one's education, may be far more hurtful than a term in jail."

Finally, this study demonstrates the multifaceted nature of disenfranchised people's socioeconomic mobility. By showing that income gains go hand in hand with stigmatization processes, the study unveils the complexities of socioeconomic mobility, which may cause both pleasure in terms of economic gains and pain due to socio-psychological distress. These findings are consistent with Merton's (1957) arguments, according to which social phenomena may have intended (or manifest) and unintended (or latent) consequences, and scholars must consider both aspects. The allegory of the favela summarizes how this bittersweet process occurs and suggests its potential generalizability to individuals experiencing socioeconomic mobility in other contexts.

Practical Implications

In addition to the theoretical contributions described, the results of this study also have two salient practical implications. First, the study presents evidence on the importance of including a new set of outcomes in evaluations about the effectiveness of training programs and policies dedicated to disenfranchised individuals. This is of utmost importance since economic gains, which are prevalently measured, might mask socio-psychological distress, which is rarely measured and has consequences for individuals' mental health and well-being. A broader evaluation of these programs that considers individuals more holistically, not only in terms of their economic production, would be beneficial.

Second, the results have essential implications for entrepreneurship among disenfranchised individuals. This career path seems to be a viable option for these individuals, as it provides further opportunities for social integration beyond the formal labor market. Moreover, these activities enable them to have an impact on their communities, helping their local peers and, consequently, their neighborhoods to develop. Entrepreneurship brings these individuals an additional sense of responsibility and fosters market and economic development in areas that the public and private sectors often neglect. Policies aiming to lower the cost of capital or even develop specific credit lines and funding for individuals living in segregated and stigmatized neighborhoods might be an opportune way forward. Such actions may contribute to the survival and scale of local enterprises and, by extension, to the economy in their communities.

LIMITATIONS, ROBUSTNESS CHECKS, AND FUTURE RESEARCH

Several limitations deserve to be mentioned. First, as is conventionally known, generalizability is an issue with field experiments. Particularly in this case, as the randomization was performed among people interested in a spot in the training program, the results are generalizable to the extent that individuals are interested in receiving training to enter the labor force. This limitation is of theoretical importance since it is unclear whether the patterns observed in this group would be similarly present in other favela dwellers who are not actively looking for labor market opportunities or among residents of other segregated and stereotyped neighborhoods that are not Brazilian favelas.

Second, the small sample size (207 individuals) prevents additional investigations of heterogeneous treatment effects conditional on participants' job type allocations at the endline (survey t_1). Despite efforts to address these issues (see Online Appendices 10 and 11), the lack of statistical power in the test makes the heterogeneous treatment effect results suggestive rather than definitive. Despite acknowledging it as a limitation, I believe this feature offers an opportunity for future studies to investigate how different job allocations affect economic and socio-psychological outcomes, particularly stigmatization.

Third, the limited sample size might also affect the quantile regression results. Despite using bootstrap techniques to estimate the relationship between income and stigma on each decile, the constrained set of individuals present in each decile (on average 23 individuals per decile) might raise concerns. Aiming to address this limitation, Online Appendix 12 presents an

additional set of quantile regression tests using quartiles (i.e., $q25$, $q50$, and $q75$), which have, on average, a more extensive set of individuals per block (on average 69 individuals per quartile). Despite slight changes in significance and magnitude, the pattern of results holds. This result is of conceptual importance to validate that income and experiences of prejudice go hand in hand as upward socioeconomic mobility occurs.

Finally, the measures of stigma and discrimination were collected through a self-reported questionnaire (Online Appendix 3). Therefore, they may represent real or perceived discrimination. While acknowledging this as a limitation, I point out that the favela stigma is the only stigma type that significantly increased after the treatment (see Online Appendix 4). Similarly, the favela stigma is the only type that significantly soared among higher-income deciles (see Online Appendix 8, Panel B). These results potentially serve as falsification tests since the actual or perceived instances of discrimination appear to have impacted exclusively the residential domain and not others. These results are also conceptually important since they suggest that in this context, non-observable status characteristics, such as residence location, seem to be a more salient source of labor market discrimination than are more easily observable characteristics, such as gender or race.

Future Research

This study presents opportunities for theoretical and empirical developments in future research. First, to the best of my knowledge this is the first study to consider both economic and socio-psychological measures of well-being among disenfranchised people living in stigmatized neighborhoods, testing the effects of an intervention on both sets of outcomes. This practice should be incorporated into future scientific studies and public policies to help stakeholders holistically comprehend the consequences of similar interventions. Moreover, this study presents the possibility of considering other socio-psychological outcome measures not measured here, such as creativity, faith, resilience, sturdiness, and happiness.

Second, future research may build on this study by aiming to fully engage and understand how and why entrepreneurship might be a good career option for disenfranchised individuals, such as favela populations, formerly incarcerated people (Hwang and Phillips, 2023), immigrants and ethnic minorities (Marinoni, 2023), and women in specific institutional settings (Thébaud, 2015), among others. Moreover, future research could directly explore how entrepreneurship as a career choice might contribute to socioeconomic development in stigmatized neighborhoods. Similarly, future studies could investigate to what extent public interventions might spark entrepreneurial activities in disenfranchised communities, consequently contributing to their development.

Finally, and perhaps most important, this study highlights the advantages of using abduction as a methodological approach in theory development, particularly in the context of pre-registered randomized field experiments. While pre-registration promotes transparency and accountability, it also potentially limits further exploration and theorization of unexpected findings. This study offers insights for researchers grappling with these constraints. It demonstrates how abduction allows for adherence to pre-registration protocols—maintaining the

integrity of the original description, data, and tests—while also facilitating the investigation of unanticipated findings and intricate relationships. Future research on the complex dimensions of socioeconomic mobility could benefit from this approach. For example, a study might pre-register a hypothesis that increased socioeconomic mobility is associated with higher socio-psychological distress. Through abduction, researchers could explore underlying reasons for this coexistence, identify behaviors indicative of socio-psychological unrest, understand their manifestations, and examine factors that may mitigate or exacerbate this relationship.

Conclusion

Investigating the causal effect of a training program on the economic and socio-psychological outcomes of disenfranchised individuals subject to residential segregation, this article engages an essential topic for the achievement of the United Nations Sustainable Development Goals, specifically the goals related to poverty eradication (SDG #1), inequality reduction (SDG #10), and sustainable cities and communities (SDG #11). The study calls for change in societal mindset when programs and policies attempt to include disenfranchised individuals in socioeconomic systems. It advocates for a more holistic and humane approach that adequately supports individuals' liberation (Freire, 1970), allowing them to achieve their full potential and moving us closer to a society in which prosperity is an attainable reality for all.

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Supplementary Material

Supplementary material for this article can be found in the Online Appendix at <http://journals.sagepub.com/doi/suppl/10.1177/00018392241240469>.

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APPENDIX

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APPENDIX 1 – Phase One Classes (Top Pictures), Convocation Ceremony (Middle Picture), and Endline (survey t₁) Data Collection (Bottom Pictures)



Note: participants' faces were blurred (de-identified) following ethical recommendations.

APPENDIX 2 – Description of Additional Methodological Procedures

Below, I detail several methodological procedures.

Stratification Rationale: Specifically, the rationale behind the stratification is the following: *Age*, to avoid a random sample with a higher concentration of older or younger participants in one group and to investigate whether the program effect was stronger among younger or older participants. *Income level*, to examine whether an initially slightly higher level of resources would heterogeneously affect the program results. *Location*, to ensure that there was a balanced number of control and treated individuals across geographic regions and to prevent any misleading results from differences in the development level of different favelas. As described in the study, the results of the stratification showed that none of the strata drive heterogeneity of the program effects. This finding helps to ensure that the program results are robust and valid for groups with these different characteristics.

Pre-registration: The pre-registration protocol was performed at the American Economic Association Registry for Randomized Controlled Trials (a.k.a. AEA RCT Registry) and can be accessed at the link < <https://www.socialscienceregistry.org/trials/2765> >. Last accessed on January 30th, 2024. The registry also contains pre-trial and post-trial information, all the legal documents signed both by *Banco da Providência* and the Rio de Janeiro Secretary of Social Assistance and Human Rights, as well as the dictionary, dataset, and main analysis code.

Additional Details on the Randomization Procedure: Note that I did not re-randomize my treatment assignment because it would have increased the final standard errors of my analysis, resulting in “more harm than good (...) on moderate sized trials” (Bruhn and McKenzie, 2009: 210).

Socio-psychological Questionnaire Pilot: Note that before the baseline data collection (survey t_0) a pilot test of the socio-psychological questionnaire was performed with ten individuals who were not part of the sample, rather from other locations in which BP operates. After iterative discussions with BP managers and social workers (hereafter enumerators), a final version of the questionnaire was defined (please, see APPENDIX 3), which contains a subset of questions used in this study. The procedure follows methodological recommendations on survey application and field experiments (The World Bank, 2000; Singleton Jr. and Straits, 2009). Moreover, the constructs follow a “rowmean” calculation to maximize the number of observations. Since participants could opt not to respond to a specific question, some statements have no response, but others, from the same construct have. Therefore, I consider the average of the answered responses.

Additional Consideration on Cronbach's Alpha Scores: A Cronbach's alpha of 0.50 is considered “poor”. However, there is some discussion as to what would be an adequate cutoff for the Cronbach's alpha measure (Lance, Butts, and Michels, 2006). Therefore, given the small number of items of the construct, I believe that, for the purposes of this research, the reliability level is sufficient.

Additional Details on the Interaction with CRASs and BP teams: Follow-up meetings with enumerators and BP managers occurred every other week during the program. Also, before baseline data collection (survey t_0) and endline data collection (survey t_1), I held from one to three additional meetings with the whole team, on top of regular individual meetings with BP managers and CRAS coordinators. The objective was to ensure everyone (enumerators, BP managers, and CRAS coordinators) understood the data collection process properly, had a clear knowledge of what was being asked, and comprehended the general objectives of the research. During these meetings, enumerators also had the chance to give suggestions, and flag possible problems they

experienced in the field. This enabled me to correct or exclude questions that were generating confusion in the participants, before defining the final questionnaire. This procedure follows experimental economics good practices (Glennerster and Takavarasha, 2013; Banerjee and Duflo, 2017).

Additional Details on Survey Applications: The surveys were applied by trained enumerators who ensured that participants understood what had been asked and recorded their answers accordingly. I adopted this process to ensure participants were not randomly choosing responses and adequately understood what was being asked. The initial version of the questionnaire had more questions; however, after a pilot with ten individuals who were similar to the experiment participants, enumerators reported difficulties in going over the whole questionnaire. Enumerators flagged questions that were generating confusion and questions that the participants were answering as “I already answered that.” Enumerators described what, in their opinion, would be the ideal interview time/questionnaire length (from 45 to 60 minutes) to make sure participants were paying attention and were consistent in their answers. Notice that the questionnaire has some “reverse” questions, which were used to verify if individuals were answering consistently. For example, see question 1 in the self-efficacy block and questions 2, 4, 6, 8, and 10 on the big-five personality traits. It is noteworthy to mention that five participants reported having an individual income higher than their household income at baseline. Since in this setting, many individuals can work occasionally, or even for only a day or a couple of hours, enumerators were instructed to probe responses but not to interfere with final answers. Thus, income variables represent what individuals literally reported during their interview surveys both at baseline and endline. I acknowledge that some of these survey procedures could have been superiorly implemented, but considering the difficulties in performing data collection in the

present context, I opted to reduce the questionnaire length and respect individual final responses in the hope of having a more accurate and precise data collection in a timely manner. I also acknowledge that this procedure might introduce some degree of noise; however, I do not expect systematic bias in the responses between groups (treated and control) and across time (baseline and endline).

Additional Details on the Endline data collection: The endline data collection (survey t_1) happened from two to three months after the end of the nine-month program. This period was planned so that participants would have enough time after the end of classes to find a job and start their business. Finally, it is important to disclose that in the endline data collection, BP gave a pair of shoes to all participants who responded to the survey, both control and treatment. This served as a non-monetary compensation for the respondents' time. I do not expect this to generate any bias either in the individuals' responses or in the comparison between groups since everyone received the same incentive.

Additional Tests with BP Regular Operations: A non-randomized comparison with the “pure private” governance structure, in which the BP offers its program without the partnership with the public organization, was also performed (as described in the pre-registration). As these analyses are non-randomized and in different locations, their statistical inference are limited. Therefore, I am not including them here.

Multiple Hypothesis Testing Robustness Checks: APPENDIX 5 brings a sequence of robustness checks considering potential problems related to Multiple Hypothesis Testing and alternative ways to estimate it (Westfall and Young, 1993; Romano and Wolf, 2005; List, Shaikh, and Xu, 2019; Clarke, 2021; List, Shaikh, and Vayalinkal, 2021). The majority of the results confirm the main results of the paper, with the exception of the “Optimism” variable, which

appears as not significant at ($p \geq 0.1$) in all tests. Since the optimism construct was not a central one for the paper argument, we claim the validity of the main results for the purpose of theory development.

Additional Details on Groups Comparison at the Baseline: According to Table 2 (in the main paper), the groups are statistically different only for two variables. The treated group has a marginally higher self-efficacy score ($p < 0.1$), and it has a slightly higher number of people living in the same household ($p < 0.05$).

Cost-Benefit Analysis: A cost-benefit analysis (a.k.a. CBA) varying the income appreciation/depreciation as well as the discount rate of a lifetime cash flow of treated versus control participants has been performed. The results suggest that the program is cost-efficient (i.e., positive Net Present Value). Due to space limitations and the lack of more refined analysis incorporating taxes and welfare implications, these analyses have been omitted.

Additional Details on Mediation and Moderation Tests: An essential interrelationship between socio-psychological outcomes and economic outcomes occurs. This suggests that it is critical to test the second-order effects of socio-psychological changes on economic outcomes. APPENDIX 12 shows a mediation test of the socio-psychological outcomes on the economic outcomes. This test indicates that increases in self-efficacy explain from 20 to 30 percent of the income gains of participants at the endline (survey t_1). APPENDIX 13 shows a moderation test of socio-psychological outcomes at the endline (survey t_1) on the relationship between the treatment and income at the endline (survey t_1).

APPENDIX 3 – Questionnaire on Self-Efficacy, Optimism, and Social Stigma

	QUESTION - From a score of 1 to 10, where 1 means “Totally Disagree” and 10 means “Totally Agree,” evaluate the following statements:	N/A Does not apply or does not want to respond	Score from 1 to 10
SELF-EFFICACY			
Q1	Compared to other people I consider myself LESS capable of accomplishing my work. (R)		
Q2	In comparison to other people, I believe that I perform some jobs better.		
Q3	In comparison with other people, I consider myself more competent to carry out my work.		
OPTIMISM			
Q4	I believe this year will be better than last year.		
Q5	In general, I try to see the good side of things, even when the situation is difficult.		
Q6	I am optimistic about my life and try to make other people optimistic as well.		
SOCIAL STIGMA			
Q7	I often suffer from prejudice because I live in a <i>favela</i> .		
Q8	I believe I have a harder time getting a job because I live in a <i>favela</i> .		
THE SHORT VERSION OF 5 BIG PERSONALITY TRAIT			
BF1	I see myself as extroverted, enthusiastic		
BF2	I see myself as critical, quarrelsome (7R)		
BF3	I see myself as dependable, self-disciplined		
BF4	I see myself as anxious, easily upset (9R)		
BF5	I see myself as open to new experiences, complex		
BF6	I see myself as reserved, quiet (1R)		
BF7	I see myself as sympathetic, warm		
BF8	I see myself as disorganized, careless (3R)		
BF9	I see myself as calm, emotionally stable		
BF10	I see myself as conventional, uncreative (5R)		

Note: “(R)” means reversed item. Other types of stigma were measured using the same structure of questions #7 and question #8. The poverty stigma had an additional question asking if participants “believe that people are prejudicial against the ones who receive *Bolsa Familia*.”

APPENDIX 4 – Treatment Effect on Alternative Stigma Aspects of Participants

As mentioned in the body of the paper, other possible types of stigma that might affect participants are related to poverty, gender, or race. As a falsification test, I tested these three types of stigma using questions similar to APPENDIX 3 questions #7 and #8 but substituting the term “favela” for terms related to poverty (e.g., “being poor or receiving *Bolsa Família*”),¹ gender (e.g., “being a woman”), and race (e.g., “being a person of color”), respectively. Internal reliability tests performed after final data collection show a Cronbach's alpha of 0.60 both for poverty and race stigmas and 0.67 for gender stigma. The estimations show a null result of the program on these potential other stigmas. Notice that the models here include each stigma type baseline (survey t_0) score.

<i>DVs =</i>	(1) Poverty Stigma in t_1	(2) Poverty Stigma in t_1	(3) Gender Stigma in t_1	(4) Gender Stigma in t_1	(5) Race Stigma in t_1	(6) Race Stigma in t_1
Treatment	0.239 (0.285)	0.0633 (0.288)	0.0852 (0.384)	-0.260 (0.362)	0.194 (0.369)	-0.0650 (0.366)
Constant	7.609** (0.215)	6.147* (2.538)	5.467** (0.275)	1.945 (2.806)	5.328** (0.246)	4.170 (3.394)
Controls	N	Y	N	Y	N	Y
<i>DV Average</i>	7.741		5.515		5.435	
Observations	207	207	206	205	207	206
R-squared	0.003	0.334	0.000	0.387	0.001	0.301

Note: Controls include randomization strata fixed effects, training period fixed effects, demographic characteristics, big five personality traits, and outcome values at the baseline (survey t_0), including the values of these additional stigmas at the baseline (survey t_0). Robust standard errors in parentheses. Socio-psychological outcomes are measured using a scale ranging from 1 to 10 points. ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$

¹ Poverty stigma had an additional question asking if participants “think people are prejudicial against the ones who receive *Bolsa Família*.”

APPENDIX 5 – Multiple Hypothesis Testing Results

I performed multiple hypothesis testing following the literature on experimental economics (Westfall and Young, 1993; Romano and Wolf, 2005; List, Shaikh, and Xu, 2019; Clarke, 2021; List, Shaikh, and Vayalinkal, 2021). The majority of the results confirm the main results of the paper with the exception of the “Optimism” construct, which appears as not significant in all tests.

Multiple Hypothesis Testing, Various Estimations

Outcomes at the endline (survey t_1)	Coeff. by Westfall- Young (1993)	Westfall- Young (1993) Unadjusted p -value	Westfall- Young (1993) Adjusted p -value	Bonferroni- Holm Adjusted p -value	Sidak-Holm Adjusted p -value	Romano- Wolf (2005) Resample p -value	Romano- Wolf (2005) p -value	Young (2019) Randomiz-C Maximum p -value	Young (2019) Randomiz-T Maximum p -value
No income, no job	-0.11	0.109	0.331	0.436	0.370	0.115	0.336	0.113	0.114
Formal job	0.04	0.371	0.596	0.743	0.641	0.367	0.597	0.387	0.378
Informal job	-0.05	0.417	0.596	0.743	0.605	0.417	0.597	0.424	0.424
Entrepreneur	0.12	0.000	0.003	0.001	0.001	0.003	0.004	0.000	0.000
Income per capita	98.65	0.001	0.006	0.011	0.010	0.001	0.007	0.002	0.001
Household income	351.21	0.000	0.003	0.003	0.003	0.001	0.004	0.000	0.000
Individual income	133.20	0.043	0.163	0.217	0.199	0.043	0.166	0.046	0.044
Self-efficacy	0.60	0.008	0.041	0.053	0.051	0.006	0.045	0.008	0.007
Optimism	0.26	0.179	0.444	0.537	0.447	0.189	0.449	0.189	0.193
Favela Stigma	1.02	0.008	0.041	0.053	0.051	0.008	0.045	0.008	0.009
Randomization p-value for Westfall-Young multiple testing of treatment significance:								0.004	0.001

Note: All models were estimated with no control variables included. All estimations used 5,000 bootstrap replications. Red cells indicate a p -value under 90% C.I., yellow cells 95% C.I., and green cells 99% C.I. White cells indicate that a corresponding p -value is above 90% C.I., i.e., non-significant (n.s.).

For more information, check David McKenzie's (2020) World Bank blog post about MHT available at: <

<https://blogs.worldbank.org/impac evaluations/overview-multiple-hypothesis-testing-commands-stata> >. Accessed on 08th February 2023.

APPENDIX 6 – Computational Text Analysis over Phase One Written Feedback

A plausible question that arises is whether the experience of stigma occurs during or after the BP training program. This is a critical consideration because, if the stigma occurs during the program, then it might suggest that the program in itself is the source of the experience. If the experience of stigma occurs after the program, then society's response is more likely the driver.

To address this question and to provide a more nuanced view of the quantitative results, the study includes qualitative analyses of participants' written feedback after Phase One of the training program. As described previously, there are three phases in the program, and Phase One is the one that focuses on socio-psychological developments. APPENDIX 1 presents pictures of the Phase One classes both in *Panuva* (top left) and *Cidade de Deus* (top right) CRAS. By the end of Phase One, participants are asked to give feedback similar to a course evaluation in which they can score aspects of the course and make written comments. Below, the image shows a scanned version of a feedback, as well as its English translation.

Example of Phase One Feedback, Original (Left) Translated (Right)

Deixe aqui seu depoimento:

Primeira e importante etapa para mim
foi última fiz um sair de uma delegacia
que comencem novamente com outras
pessoas foi uma mais experiencia de trabalhar
com outros e fazer pilos outros objetivos
necessariamente.

Cada passo foi muito importante para
meu trabalho se aplicar e trabalhar mais
uma mais com mais e também saber
o nosso direito como ser humano
na sociedade e para meus estudos
também para fazer que como se
tivera faculdade um dos pontos que se
concentra em de pessoas que vivem
como vivem a diferença
espero que esse projeto possa ajudar
também outras pessoas com a ajuda da
ajudada e sociedade atropada.

Desejamos que você tenha sucesso em seu plano!
Esta folha é para ser entregue a assistente social

English translation:

"Well, this first stage for me was great, it made me get out of a depression to communicate again with other people, to have new hope of fulfilling my dreams and pursuing my goals again. Each step was very important because it is not enough to earn the cash but to know ourselves better and also to know our rights as a human being in society. I could also put my fears out, which was like dropping down the weights I was carrying. Even the people who live with me saw the difference. I hope that this project can help as many other people as I could be helped and guided. Thanks."

-- Participant "D" feedback after Phase One.

Note: Banco da Providência archives. The files were de-identified, scanned, and digitized by the author.

The Computational Text Analysis Methodology

Computational text analysis was applied to these written pieces of feedback. This method has been used in the management literature to understand patterns in innovation, cognition, and firm capabilities (Kaplan, 2008, 2011; Kaplan and Vakili, 2015). Following this trend, a suitable text analysis for the current research project is sentiment analysis (Liu, 2012; Henisz, Dorobantu, and Narthey, 2014; Dorobantu, Henisz, and Narthey, 2017; Choudhury et al., 2019). Sentiment analysis identifies the emotions and sentiments (e.g., negative-positive, conflict-cooperation) implicitly present in a set of written documents (a.k.a. corpora). As the project aims to evaluate the training program's impact on socio-psychological outcomes, the sentiment analysis provides insights on the mechanisms through which the program increases self-efficacy and optimism among participants. This analysis also helps to identify when experiences of stigma emerge.

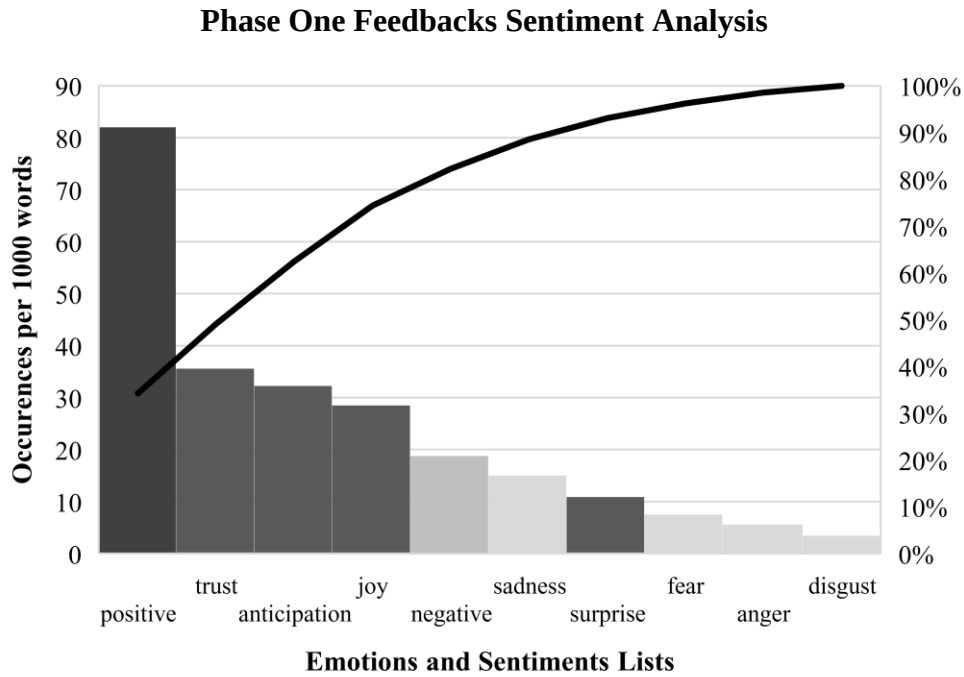
A total of 71 pieces of Phase One feedback were digitized for the sentiment analysis. A Python[®] code parsed each word within each piece of feedback. A lexicon called the NRC Word-Emotion Association Lexicon (a.k.a. EmoLex) was used to classify each word according to the evoked sentiment (Mohammad and Turney, 2013). EmoLex first classifies words as positive or negative and second across eight different sentiments, of which four are positive (i.e., anticipation, joy, surprise, and trust), and four are negative (i.e., anger, disgust, fear, and sadness). As the pieces of feedback were written in the Brazilian-Portuguese language, a translated version of EmoLex was employed (Mohammad and Turney, 2013). Finally, the study calculated the feedback $Sentiment Score_{wsi}$ considering each word w , and its sentiment s , for each individual piece of feedback i using the formula:²

² Note that the sentiment score formula considers text length (in the denominator), which correct for possible biases driven from long feedbacks.

$$Sentiment\ Score_{wsi} = \left[\frac{\sum_{w=1}^W \mathbf{1}\{Word\ Sentiment\ Count_{wsi}\}}{\sum_{w=1}^W w_i} \right] * 1000$$

Qualitative Results and Discussion

The figure below shows the average sentiment scores across all pieces of feedback for each possible sentiment s . On average, positive sentiments (in dark gray colors) are more common than negative ones (in light gray colors). This corroborates the findings of the quantitative experimental analysis of an increase in self-efficacy and optimism after the treatment.

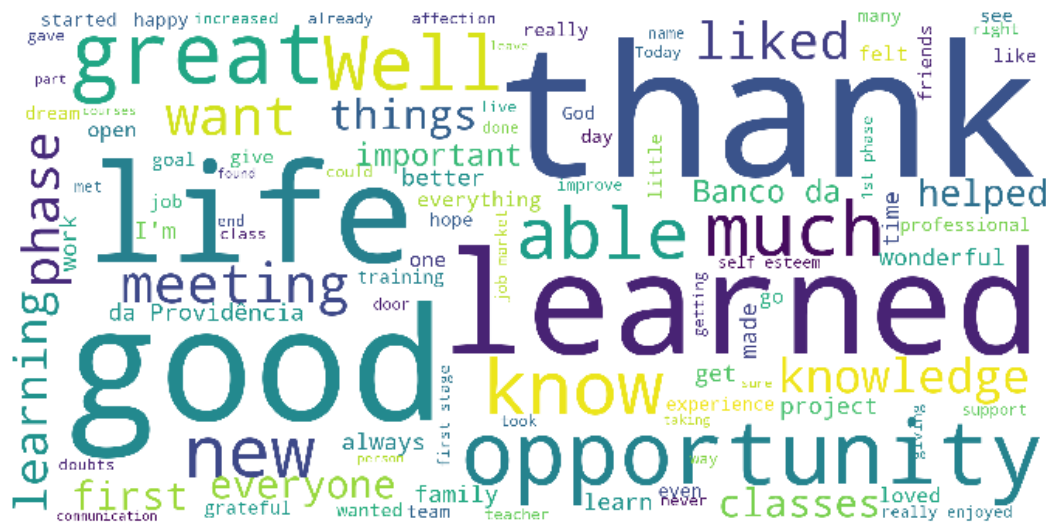


Note: The figure represents the average score for ten different emotions and sentiments defined at NRC Word-Emotion Association Lexicon (EmoLex) for the Portuguese language (Mohammad and Turney, 2010, 2013). Dark gray represents positive emotions and sentiments; light gray represents negative emotions and sentiments.

Following a recent call for visual analysis of text data in management (Ertug et al., 2018), the study includes a visual representation of the most common words in the body of feedback. Specifically, image below presents a “word cloud” of the 100 most common words in the feedback. To generate the word cloud, the code translated the contents to English, dropping stop-words, course names, and personal names (usually the names of the course instructors). The size of each

word represents the frequency of its occurrence, i.e., the bigger the size, the more frequently the word appears.³ The word cloud shows that words such as “great,” “thank,” “life,” “good,” “learned,” and “opportunity” are among the most common. In general, they map to the sentiment analysis as well as to the quantitative experimental results.

Phase One Feedbacks “Word Cloud”



Note: 100 most common words present in the participants' written feedback. This version of the word cloud presents the English-translated content from the Phase One feedbacks. Inspired by Ertug et al. (2018).

To address when the experiences of stigma occurred, the study estimates whether the participant's sentiment scores on pieces of Phase One feedback are correlated with their final socio-psychological outcomes, which occurred two to three months after the end of the program (endline survey at t1). This analysis is informative about whether Phase One influenced perceived experiences of stigma.

The table below shows that the negative sentiment arising in the post-Phase-One feedback was not correlated with final outcomes. Similarly, positive sentiment was not correlated with experiences of stigma. This stands in contrast with the positive correlation reported in Models (2)

³ This version of the word cloud presents the English translated content from the Phase One feedbacks.

and (4) between participants' positive sentiments immediately after Phase One with both their final self-efficacy score ($p < 0.1$) and optimism score ($p < 0.05$), respectively.

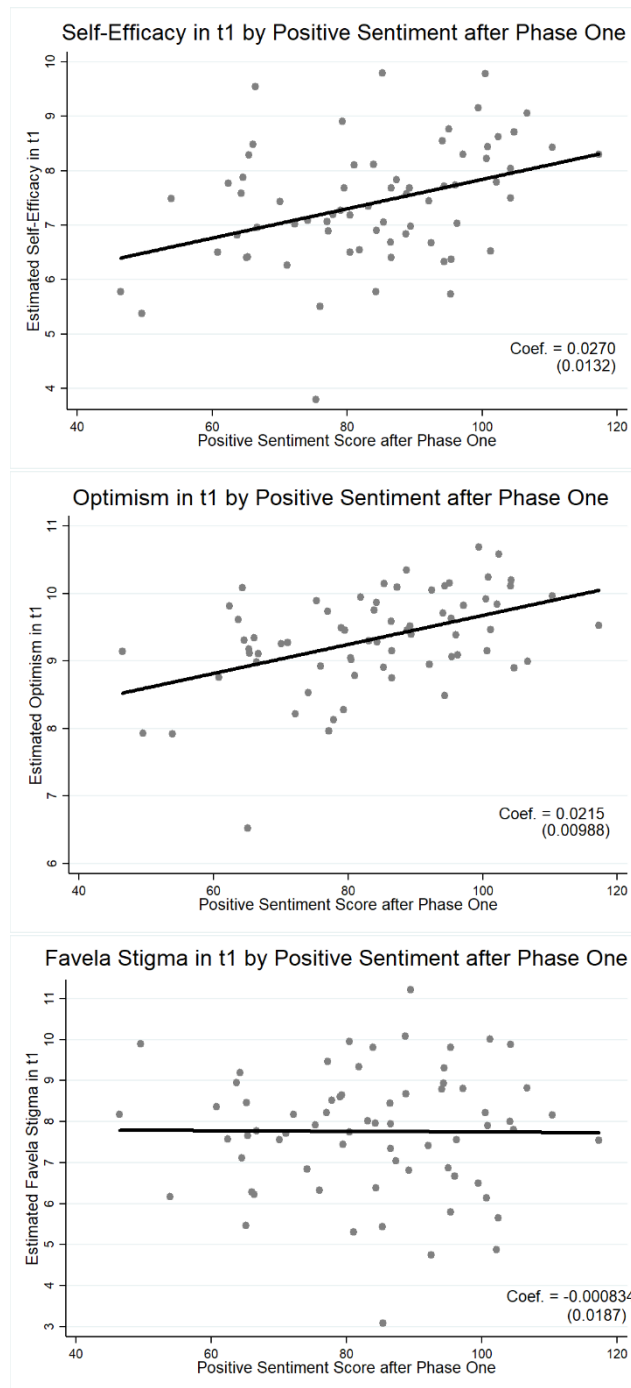
Correlation Between Phase One Feedback Positive and Negative Emotion Scores and Final Outcomes (survey t_1)

DVs =	(1) Self- Efficacy in t_1	(2) Self- Efficacy in t_1	(3) Optimism in t_1	(4) Optimism in t_1	(5) Favela Stigma in t_1	(6) Favela Stigma in t_1
Positive Emotions	0.0144+ (0.00812)	0.0270+ (0.0132)	0.0110+ (0.00578)	0.0215* (0.00988)	-0.0107 (0.0134)	-0.000834 (0.0187)
Negative Emotions	0.00711 (0.0514)	0.00554 (0.0402)	0.00132 (0.0318)	0.0104 (0.0418)	0.0108 (0.0567)	-0.0434 (0.0701)
Constant	7.516** (0.709)	0.705 (6.516)	7.983** (0.890)	1.104 (4.128)	7.523** (1.241)	18.42* (8.010)
Controls	N	Y	N	Y	N	Y
Observations	68	68	68	68	68	68
R-squared	0.259	0.658	0.238	0.710	0.181	0.687

Note: All results are controlled by randomization strata and training period. Additional controls include demographic characteristics, big five personality traits, and outcome values at the baseline (survey t_0). Also, estimations include controls for each feedback score on the eight sentiments evaluated by EmoLex (Mohammad and Turney, 2010, 2013). Robust standard errors in parentheses ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$

The results suggest that Phase One has an effect on self-efficacy and optimism but no effect on experiences of stigma. This correlation strengthens the argument that experiences of stigma increase after the training program ends as a response to deleterious interactions during job or supply searches. In other words, the null effect on the correlation between positive emotions after Phase One and stigma at the endline (survey t_1) reinforces the “getting out of the ghetto” interpretation and the predictions of the *diversity and inclusion* literature. The figure below shows the full controlled binned scatterplot correlations (Starr and Goldfarb, 2020).

Correlation Between Phase One Feedback Positive Sentiment Scores and Final Outcomes (survey t₁)



Note: Non-parametric binned scatterplot estimations (Stepner, 2013; Starr and Goldfarb, 2020). The estimations include controls for randomization strata, training period, demographic characteristics, big five personality traits, and outcome values at the baseline (survey t₀). Also, estimations include controls for each feedback score on the eight sentiments evaluated by EmoLex (Mohammad and Turney, 2010, 2013). Graphs' y-axes are on different scales.

Limitations of the Computational Text Analysis

This qualitative analysis of Phase One feedback illuminates the possible mechanisms driving the quantitative experimental results. However, three main limitations should be acknowledged. First, the study only has feedback from the treated group, not the control; therefore, it is not possible to make causal inferences.⁴ Second, the content of the feedback at the end of Phase One may be associated with more positive feelings than at the baseline (survey t_1) because participants provide it during their enrolment period in the program.⁵ Finally, the study includes information on the reported experiences of individuals rather than on observed social discrimination. Future research is needed to identify mechanisms of stigma definitely, such as through audit studies using fictitious resumés (Bertrand and Mullainathan, 2004; Kang et al., 2016) to expose discriminatory practices of employers towards favela dwellers.

⁴ Unfortunately, not all participants wrote feedback after Phase One. From a total of 123 participants enrolled in the training module of the project, only 71 participants filled in the feedback form (about 58 percent). Also, given the attrition rate, three of the participants who filled in the feedback form did not follow-up in the baseline (survey t_1). Thus, the final estimations presented have a total of 68 observations included, which constitutes about 60 percent of the total number of participants who answered the baseline (survey t_1). Finally, there are several reasons a participant might not have filled in the Phase One feedback form, the most common are cases in which participant missed that class, did not identify themselves in the form, and/or returned the form in blank.

⁵ To double-check the qualitative results found in the after Phase One feedbacks' sentiment analysis, I repeated the same procedure using the convocation speeches performed by participants immediately after they conclude the whole program. As the convocation speeches were performed by class type, depending on the training participants had, I have a reduced number of corpora ($n = 30$) which are not individualized, rather by class. Similar to the feedback analysis, I observe a greater presence of positive sentiments over negative ones. Also, the word cloud denotes words as “dream”, “good”, “know”, “learning”, “life”, and “thank”; as some of the most representative ones. Therefore, this robustness check enables me to conclude that qualitative evidence reinforces the effect of the program on socio-psychological outcomes as described by the quantitative and qualitative results. These analyses have been omitted due to space limitations.

APPENDIX 7 – OLS Biases and Quantile Regression Alternative

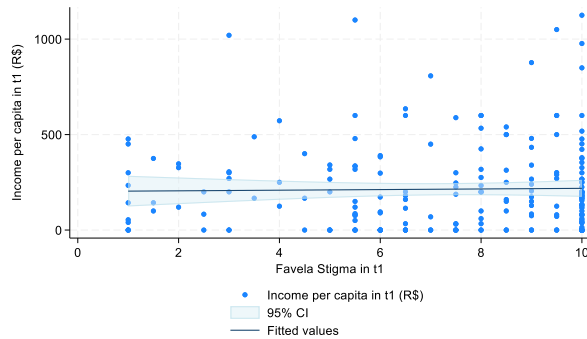
This appendix aims to clarify why the quantile regressions presented in the main paper, Table 7 and Figure 5, are a superior approach than an OLS to demonstrating the paradox of inclusion in a discriminatory system by showing the association between income per capita in t_1 and favela stigma in t_1 . To do so, several OLS assumptions will be tested using the data to show Gauss-Markov theorem is violated, i.e., OLS estimators are not BLUE (a.k.a. Best Linear Unbiased Estimators). Notice that I will perform two tests one following equation (A) and another following equation (B).

$$Incomepercap_{i_{t1}} = \alpha + \beta FavelaStigma_{i_{t1}} + e_i \quad (A)$$

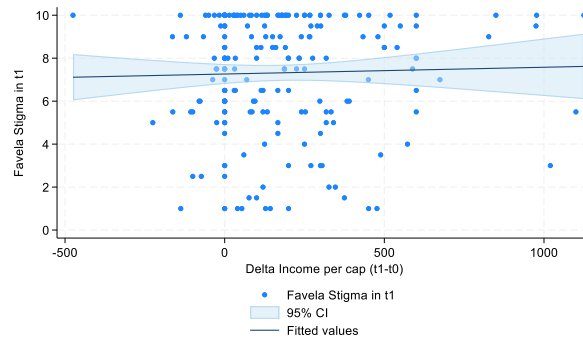
$$FavelaStigma_{i_{t1}} = \gamma + \rho \Delta Incomepercap_{i_{t1-t0}} + \epsilon_i \quad (B)$$

Linearity. OLS assumes a linear relationship between dependent and independent variables, such that: $y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + e$. The simplest way to test for linearity is through a scatter plot and or a binned scatter plot.

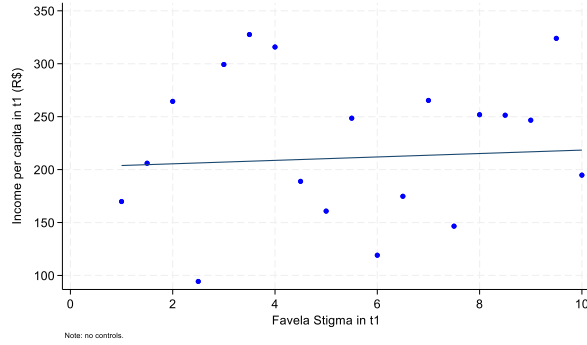
Panel (A): Scatter plot with linefitci



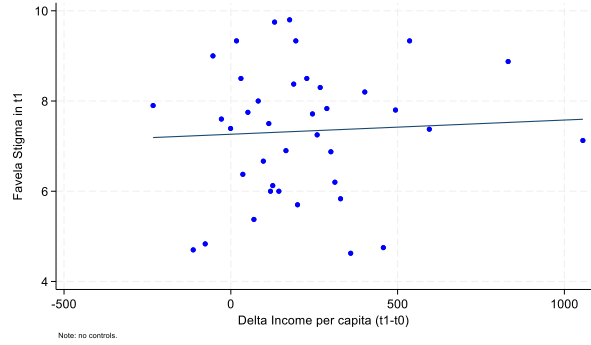
Panel (B): Scatter plot with linefitci



Panel (A): Binned scatter plot with linefit



Panel (B): Binned scatter plot with linefit



The graphs in Panel (A) and Panel (B) seem to suggest a null relationship between the variables, which is also observed in the mediation text (please see APPENDIX 12) and the moderation tests (please, see APPENDIX 13). However, and more importantly, for the purposes of this analysis, the graphs seem to raise questions about the linearity of the relationships given the data distribution.

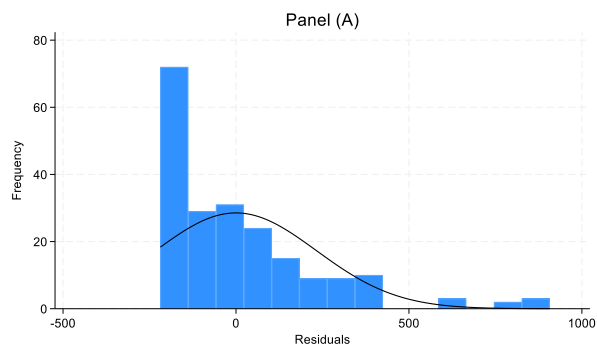
Normality. OLS models assume error terms (a.k.a. residuals) are normally distributed. Therefore, additional tests should check whether the errors are normally distributed. $e_i \sim \epsilon_i \sim N(0, \sigma^2)$.

The results from the Skewness and Kurtosis test for normality reject the null hypothesis that residuals are normally distributed both in Panel (A) and Panel (B). Moreover, the results from Jarque Bera test for normality also reject the null hypothesis, which also violates the normal distribution of the error terms. Finally, we can visually check the distribution which are displayed in the histograms below.

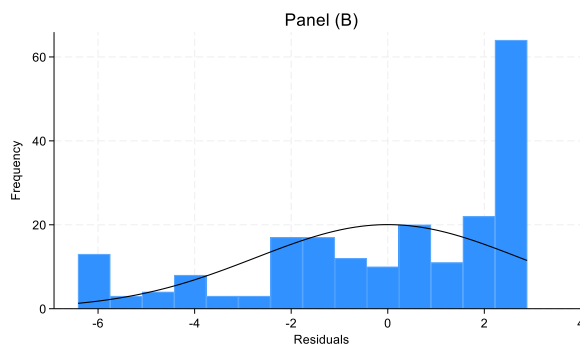
<i>DVs</i> =	Panel (A) Income per capita in t1	Panel (B) Favela Stigma in t1
Favela Stigma in t1	1.618 (5.586)	
Delta Income per capita (t1-t0)		0.000315 (0.000765)
Constant	202.2** (42.42)	7.264** (0.234)
Controls	N	N
Normality		
Skewness and kurtosis test for normality (joint test)	0.000	0.000
Jarque-Bera normality test	0.000	0.000
Homoscedasticity		
Breusch-Pagan	0.330	0.924
White's test	0.801	0.949
Observations	207	207
R-squared	0.000	0.001

Note: Robust standard errors in parentheses. ** p<0.01, * p<0.05, + p<0.1

Panel (A): Residuals Distribution



Panel (B): Residuals Distribution



Homoscedasticity. Together with the normality assumption, OLS assumes the error terms to be homoscedastic, i.e., the variance of the residuals is constant. $e_i \sim \epsilon_i \sim N(0, \sigma^2)$. Results from

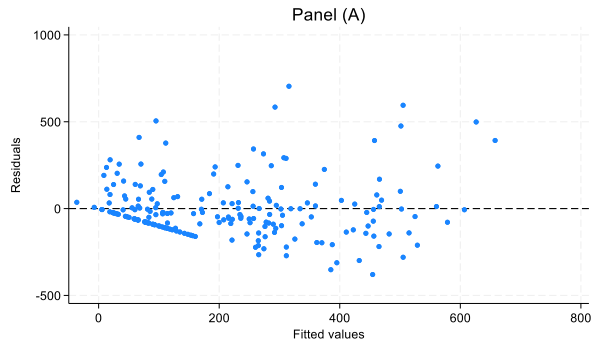
Breusch-Pagan and White's test confirms do not reject the null hypothesis, which confirms the constant behavior of the residuals. However, in a controlled model, we have both non-normality and heteroscedasticity of the error terms according to the Breusch Pagan test.

We can also visualize the heteroscedasticity of residuals in the “Residuals-versus-fitted” plots, which show the inconstant behavior of the error terms.

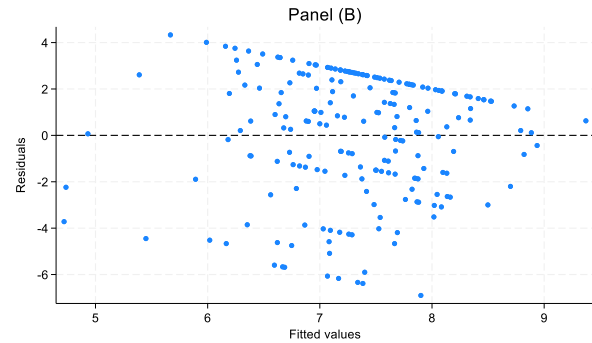
<i>DVs =</i>	Panel (A) Income per capita in t1	Panel (B) Favela Stigma in t1
Favela Stigma in t1	1.693 (4.722)	
Delta Income per capita (t1-t0)		0.000404 (0.00113)
Constant	175.6+ (98.61)	5.801** (1.476)
Controls	Y	Y
Normality		
Skewness and kurtosis test for normality (joint test)	0.000	0.001
Jarque-Bera normality test	0.000	0.000
Homoscedasticity		
Breusch-Pagan	0.000	0.003
White's test	0.099	0.702
Observations	207	207
R-squared	0.463	0.076

Note: Models control for job type allocation at the endline (survey t₁), demographic characteristics, outcome values at the baseline (survey t₀). Robust Standard errors in parentheses. ** p<0.01, * p<0.05, + p<0.1

Panel (A): “Residuals-versus-fitted”



Panel (B): “Residuals-versus-fitted”



These results show the OLS is not BLUE, and therefore, it is not the best way to estimate the association between income per capita and the favela stigma, either following model (A) or model (B). An alternative functional form that emerges as an interesting option is the quantile regression. Different from OLS, quantile regressions do not assume linearity, or normality and homoscedasticity of the residuals. Moreover, quantile regression allows us to evaluate the relationship of the two variables at different moments in the distribution of the dependent variable. Research has shown the benefit of quantile regressions in program evaluation studies as a way to verify heterogeneity of treatment by quantile of outcome (Firpo, 2007). Therefore, we applied the quantile regression as a way to test the multifaceted nature of socioeconomic mobility among individuals included in a discriminatory system, without running the biases that linear models.

APPENDIX 8 – Additional Stigma Outcomes by Decile of Income per capita in t₁ (R\$)

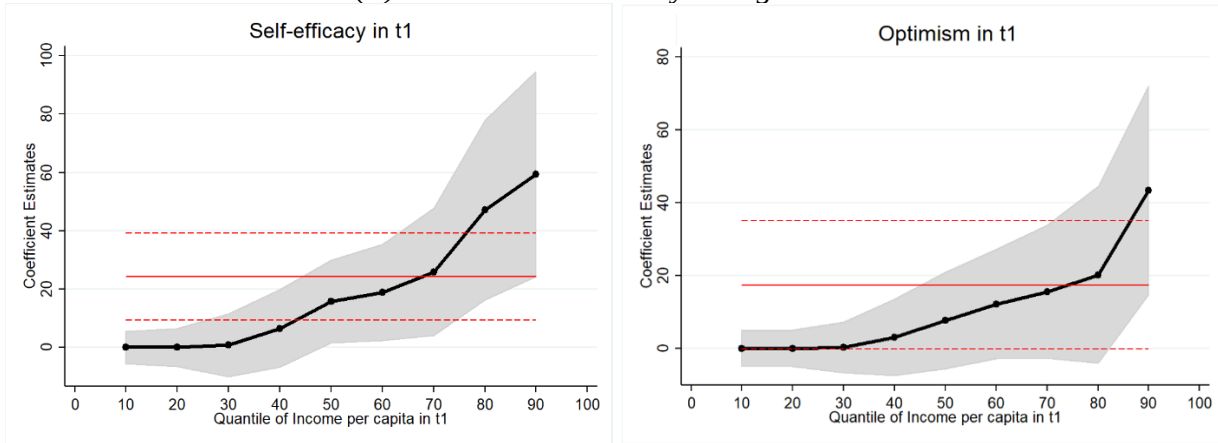
I further test for the paradox of inclusion in a discriminatory system through a quantile regression model (Koenker and Bassett, 1978; Abrevaya, 2001; Koenker and Hallock, 2001; Fitzenberger and Wilke, 2006). This analysis enables me to evaluate how stigma operates conditionally on different quantiles of income per capita in t₁ (R\$). As we can see, the additional stigma types (poverty, gender, and race) do not vary by income decile. This suggests that the favela stigma is indeed the only one increasing with higher economic prosperity.

	(0) OLS	(1) q10	(2) q20	(3) q30	(4) q40	(5) q50	(6) q60	(7) q70	(8) q80	(9) q90
Panel (A): Additional Socio-Psychological Outcomes										
Self-efficacy in t ₁	24.28** (7.129)	0 (2.762)	0.00474 (3.289)	0.790 (5.413)	6.567 (6.692)	15.76* (7.136)	18.92* (8.351)	25.92* (11.04)	47.11** (15.63)	59.42** (17.80)
Constant	36.35 (93.82)	0 (56.41)	2.255 (64.44)	7.740 (78.87)	21.49 (98.94)	80.25 (105.4)	126.9 (112.8)	17.14 (130.1)	-1.274 (166.7)	-21.56 (212.7)
Optimism in t ₁	17.39** (6.556)	0 (2.473)	0.0189 (2.534)	0.246 (3.493)	3.051 (5.278)	7.672 (6.732)	12.20 (7.596)	15.58+ (9.252)	20.15 (12.30)	43.41** (14.52)
Constant	20.05 (118.7)	0 (54.09)	1.821 (58.80)	15.61 (79.55)	-10.10 (102.4)	63.46 (120.2)	21.08 (138.0)	-11.06 (168.9)	-9.040 (216.4)	-136.0 (264.8)
Panel (B): Alternative Stigma Outcomes										
Poverty Stigma in t ₁	0.664 (5.224)	0 (2.164)	0.0150 (2.032)	0.136 (2.724)	-0.163 (4.098)	-2.937 (5.530)	-1.058 (6.839)	4.814 (8.734)	8.735 (11.80)	17.22 (13.87)
Constant	181.0+ (96.1)	0 (51.92)	1.885 (52.84)	16.81 (67.37)	43.87 (84.86)	153.5 (100.7)	151.1 (119.8)	117.2 (145.8)	65.96 (199.2)	280.1 (259.8)
Gender Stigma in t ₁	-2.116 (4.980)	0 (2.433)	-0.227 (2.109)	-0.0709 (2.841)	-1.693 (3.865)	-5.048 (4.381)	-4.895 (5.280)	1.482 (6.366)	2.725 (8.443)	7.832 (11.15)
Constant	195.6* (96.25)	14.29 (54.31)	17.44 (52.77)	19.00 (67.42)	88.15 (80.40)	124.5 (91.85)	167.7+ (100.6)	165.6 (128.5)	188.8 (190.2)	402.2 (265.4)
Race Stigma in t ₁	-5.990 (5.178)	0 (2.233)	0.00360 (1.987)	-0.0287 (2.778)	-0.630 (3.771)	-5.165 (4.732)	-3.959 (5.928)	-0.340 (7.433)	-1.139 (9.287)	-9.356 (11.05)
Constant	205.4* (97.01)	0 (45.61)	2.326 (54.86)	18.49 (71.32)	43.81 (86.50)	132.5 (95.05)	180.7 (111.2)	138.0 (136.5)	178.4 (179.1)	505.5* (242.6)
Controls	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Observations	207	207	207	207	207	207	207	207	207	207
Bootstrap Reps.	N/A	500	500	500	500	500	500	500	500	500

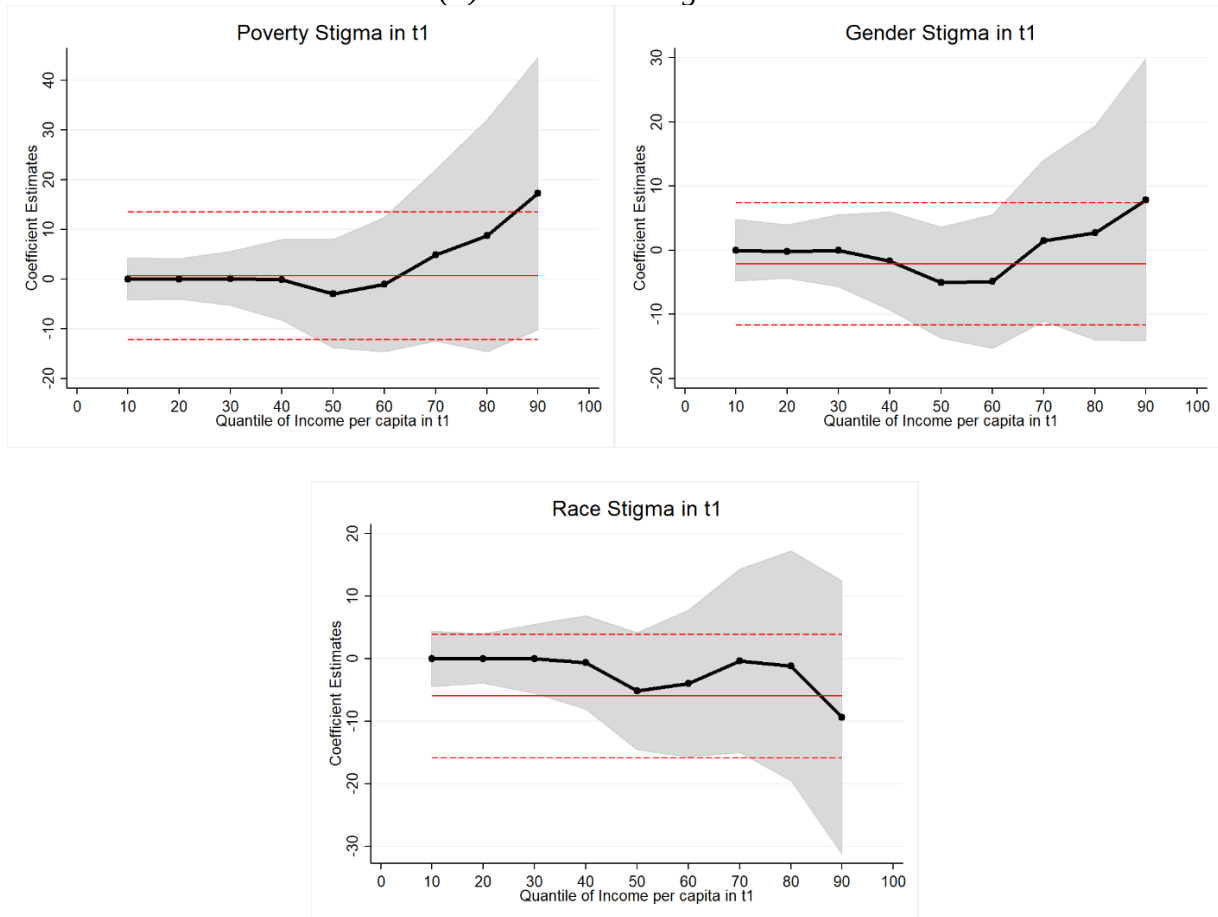
Note: all results are controlled by individual-level characteristics, namely: job allocation at the endline (survey t₁), income level at the baseline (survey t₀), and demographic characteristics at the baseline (survey t₀). Socio-psychological and personality trait baseline controls were not included to avoid “bad control” issues, given their strong intertemporal correlation. Models estimated using 500 bootstrap replications. Robust standard errors in parentheses. ** p<0.01, * p<0.05, + p<0.1

Visual Analysis of Additional Stigma Outcomes by Decile of Income per capita in t₁ (R\$)

Panel (A): Additional Socio-Psychological Outcomes



Panel (B): Alternative Stigma Outcomes



Note: Graphs' y-axes are on different scales. Point estimates (solid line) and 95% C.I. (dashed line, shaded area), red (OLS), black (Quantile Regression). All results are controlled by individual-level characteristics, namely: job allocation at the endline (survey t₁), income level at the baseline (survey t₀), and demographic characteristics at the baseline (survey t₀). Socio-psychological and personality trait baseline controls were not included to avoid “bad control” issues, given their strong intertemporal correlation. Models estimated using 500 bootstrap replications. Gender Stigma at the endline had 206 observations.

APPENDIX 9 – Further Explanations on the Allegory of the Favela

An “allegory” is defined as an artistic device used to represent a particular moral or political idea.⁶ Inspired by Plato's “Allegory of the Cave” in Book VII of “The Republic” (Bloom, 1968: 193–220), which describes the dialogue between Socrates and Glaucon, the paper performs a contemporary rereading of this dialogue to explain and generalize the study's findings. Similarly to the ancient allegory in which the image of a prisoner breaking their chains on the wall and escaping the cave represents us all while pursuing *education* to learn more about the world, in the present allegory, favela individuals getting out of their communities also represent us all in this very same pursuit. Notice that the etymology of the verb “to educate” comes from the Latin “*educat-*” which means to “led out.” This meaning applies strikingly to the “led out” process from the cave and from the favelas.

Based on the context and results presented in the study, this section details the three stages of “The Allegory of the Favela,” namely: (i) the initial misperception of reality, (ii) the pain and pleasure experienced after being freed from the initial ignorance about reality, and (iii) the potential for liberating others. The inspiration for these three phases came from Plato's “Allegory of the Cave” (Bloom, 1968: 193–220). The first stage in Plato's “allegory of the cave” refers to the period when the prisoners are chained inside the cave and face a wall where shadows are projected. Since the shadows are the only thing the prisoners know, they mistakenly consider them the reality. The second stage in Plato's “Allegory of the Cave” alludes to the moment a prisoner is freed from their chains and can finally see the world outside of the cave. This period is a combination of confusion and enlightenment. Confusion because the prisoner is blinded by the sunlight and cannot fully understand what they are seeing. Enlightenment because the prisoner eventually realizes their

⁶ Cambridge Dictionary. Available at < <https://dictionary.cambridge.org/us/dictionary/english/allegory> >. Accessed on September 13th, 2013.

initial ignorance as the actual reality of the world is revealed. The third stage in Plato's "Allegory of the Cave" refers to the phase in which the prisoner returns to the cave aiming to educate the other prisoners about the true reality of the world.

The Allegory of the Favela, Stage I: The Initial Misperception of Reality

This first stage is *ex ante* to the training program. Before going through a training program, individuals living in stigmatized neighborhoods, such as favelas, are not fully aware of their social status since they typically do not have an outside favela reference point. Moreover, these individuals are not entirely conscious of the true extent of their stigmatization by discriminatory societal systems beyond favelas. Since their daily activities are largely contained within favelas, their interactions happen among relatively comparable social status peers, and their experiences of stigma and discrimination are not acute. This socio-psychological "comfort zone" comes at the cost of underdevelopment in the educational and labor arenas. In other words, the "bliss of ignorance" about their relative status and stigmatization is accompanied by a lack of opportunities for cultural, social, and economic advancement.

The Allegory of the Favela, Stage II: The Pain and Pleasure after Being Liberated from the Initial Ignorance about Reality

This second stage is immediately *ex post* to the training program. After going through the program, favela residents develop their technical skills related to the professions they intend to perform. Concomitantly, they build their socio-psychological skills, particularly feeling more self-effective and optimistic about their potential. This educational process helps them to "break the chains" of disenfranchisement since they feel technically and psychologically prepared to enter the labor force. With this new toolkit of abilities, the treated individuals seek job opportunities and suppliers beyond the favela borders. In doing so, they realize their initial ignorance about their relative social status, which is considerably undervalued compared to their outside favela

counterparts. Eventually, they enter the labor force and increase their income levels. Yet, the actual reality of the discriminatory societal system in which they have been recently included is gradually revealed as they receive recurrently negative feedback from potential employers and suppliers who classify them as unfit, untrustworthy, or unsafe.

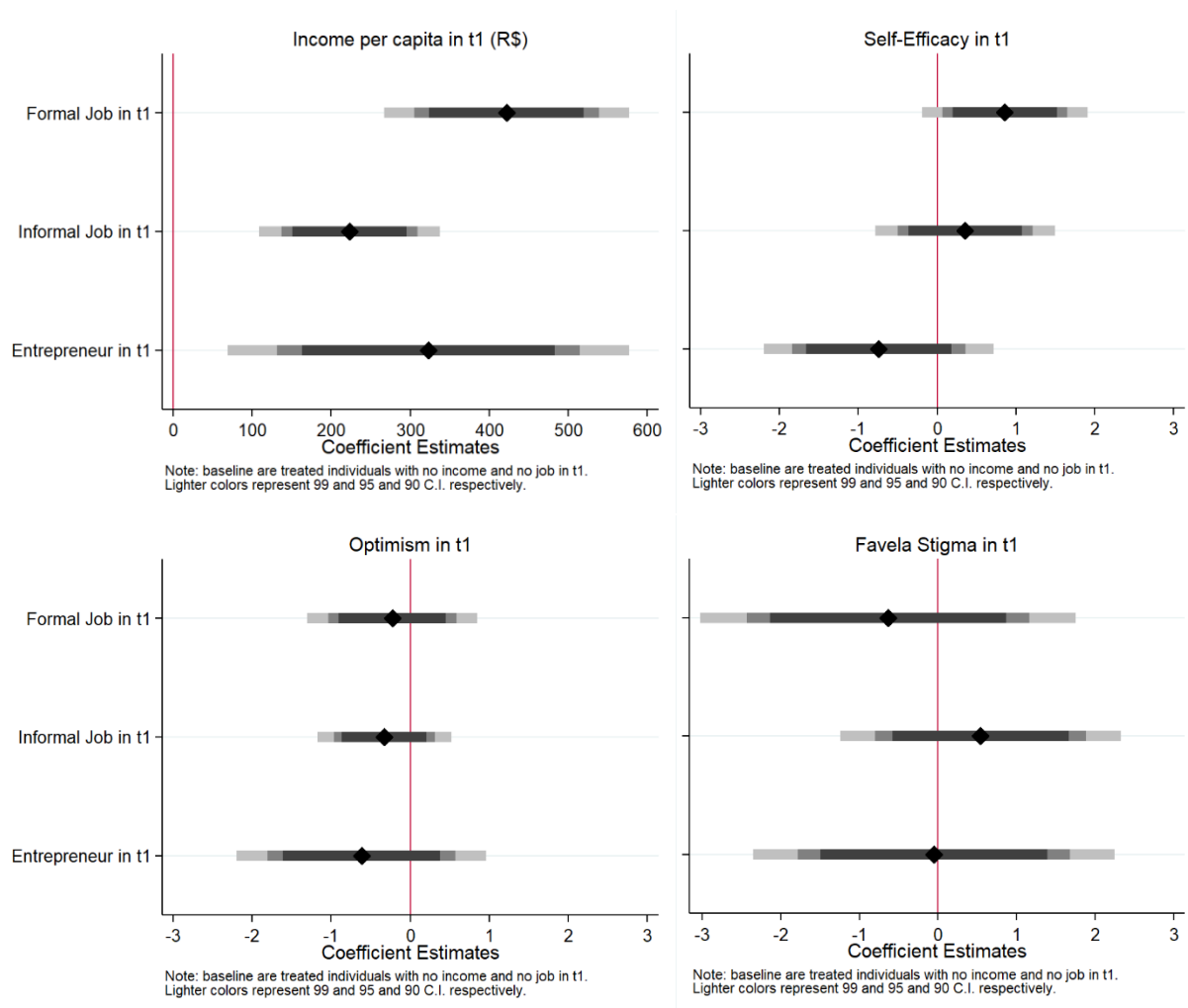
Notably, this is a bittersweet process as it brings them the satisfaction of socioeconomic mobility and its associated economic gains, but also the frustration of realizing their devalued social status and the blatant realization of the societal prejudice against them. Furthermore, individuals with more extraordinary socioeconomic prosperity (higher income levels) report even greater experiences of discrimination, given the increased relative social distance between them and their new peers.

The Allegory of the Favela, Stage III: The Potential for Liberating Others

This third stage represents the implications of the training program in the medium to long term. As the formal labor market does not significantly absorb the treated individuals, they opt to become entrepreneurs as a backup plan, which involves developing their businesses within the favelas. In doing so, these individuals bring their experiences and knowledge about the reality outside to their local communities. These entrepreneurs engage with the community on a deeper level since they know it from within. Interacting with local customers and hiring other favela residents may have a second-order effect on the economic development of the entire community. This process has the potential to liberate others from economic disenfranchisement while, to some extent, safeguarding their peers against direct interaction with discriminatory systems outside the favelas.

APPENDIX 10 – Heterogeneity Analysis by Job Allocation at the Endline (survey t₁)

Different job allocations might heterogeneously influence participants' income and socio-psychological outcomes at the endline (survey t₁). The analysis over the treated group (114 individuals) in all job allocation categories – formal, informal, and entrepreneurs – confirms this heterogeneity at the endline (survey t₁). This result reinforces the positive effect of the program on economic performance and the performance differences across job types. However, the analyses on socio-psychological outcomes show that all job categories are statistically similar, with only a higher level of self-efficacy at the endline (survey t₁) for formally employed individuals at the endline (survey t₁) ($p < 0.05$). The coefficient plots and regression table present these results.



<i>DVs =</i>	(1) Income per capita in t ₁ (R\$)	(2) Self-Efficacy in t ₁	(3) Optimism in t ₁	(4) Favela Stigma in t ₁
Formal Job in t₁	421.9** (58.61)	0.858* (0.398)	-0.226 (0.407)	-0.635 (0.904)
Informal Job in t₁	223.3** (43.29)	0.356 (0.431)	-0.326 (0.321)	0.542 (0.675)
Entrepreneur in t₁	323.3** (96.07)	-0.740 (0.552)	-0.614 (0.597)	-0.0526 (0.870)
Constant	149.6 (376.5)	5.002+ (2.982)	4.635* (2.129)	5.057 (3.901)
Controls	Y	Y	Y	Y
Observations	114	114	114	114
R-squared	0.626	0.459	0.421	0.355
<i>Joint Sig. F-test</i>	<i>20.7</i>	<i>3.1</i>	<i>0.7</i>	<i>0.5</i>
<i>p-value of F-test</i>	<i>0.000</i>	<i>0.034</i>	<i>0.534</i>	<i>0.698</i>

Note: Models consider only treated individuals (n = 114). The excluded dummy (reference group) is composed of individuals with “no income, no job condition” at the endline (survey t₁). All results are controlled by randomization strata and training period. Additional controls include demographic characteristics, big five personality traits, and outcome values at the baseline (survey t₀). Robust standard errors in parentheses. ** p<0.01, * p<0.05, + p<0.1

APPENDIX 11 – Evolution of the Entrepreneurs

To what extent do the effects of the program depreciate or appreciate over time? To this end, a follow-up survey was performed one year after the end of the program (survey t_2) only with the participants that declared themselves as entrepreneurs at the endline (survey t_1). I focus on this subset of participants because entrepreneurship was one of the primary outcomes of the program. Also, I considered the challenges related to re-accessing all participants after one year from the program's end. Moreover, the option for the entrepreneurs enabled me to explore additional information on their business ventures, which might be interesting to analyze.

In terms of *economic outcomes*, 61 percent of the entrepreneurs are still operating their businesses after one year from the end of the program (survey t_2). Despite an apparent high failure rate (39 percent), among the individuals who left the entrepreneurship activity, 60 percent did so to enroll in the formal labor market, which is usually associated with more income and stability. In any case, analyzing the evolution of the entrepreneurs, we see that income levels remained statistically similar to the ones at the end of the program, and in the case of individual income, it even increased ($p < 0.05$). This result suggests that the potential effect of the program in terms of income is positive or at least neutral (see figures on the “income evolution among entrepreneurs”).

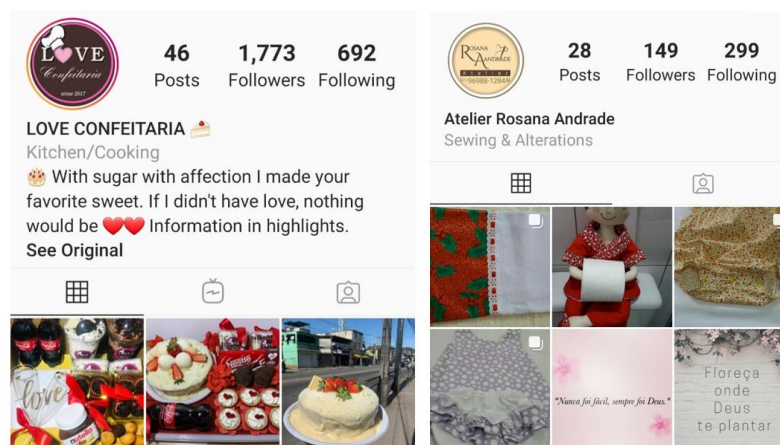
Moreover, in terms of *socio-psychological outcomes*, the results are complex. After one year from the end of the program, entrepreneurs significantly decrease their levels of self-efficacy ($p < 0.05$). This might suggest that after some time in the market or after failing as an entrepreneur, individuals reduce their perception of their capabilities. Interestingly, the experiences of stigma decreased in the same period ($p < 0.05$), which might reflect that individuals are either facing less prejudice after building trust through recurrent interactions in the market (Jones et al., 2014), or they developed resilience against this type of deleterious social interactions. Finally, no statistical

effect is captured in optimism (see figures on the “socio-psychological evolution among entrepreneurs”).

As mentioned previously, the follow-up with the entrepreneurs explored second-order effects that these individuals have on their communities. In this sense, an extra block of questions was asked to the individuals that remained in entrepreneurial activities. These questions tried to map specific aspects of their business ventures. For example, on average, the entrepreneurs invested R\$ 500 (U\$ 125) in their business, mainly buying capital such as machines and tools; 25 percent formalized their business and hired someone from the favela to work with them; 83 percent sell their products or services in the favela; and 20 percent buy inputs in the favela. Moreover, 62.5 percent described the use of social media, such as Instagram® and WhatsApp®, as a marketing strategy (see illustration below); and 37.5 percent of the entrepreneurs moved out of the *Bolsa Família* program.

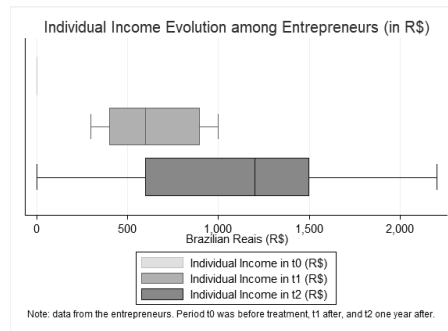
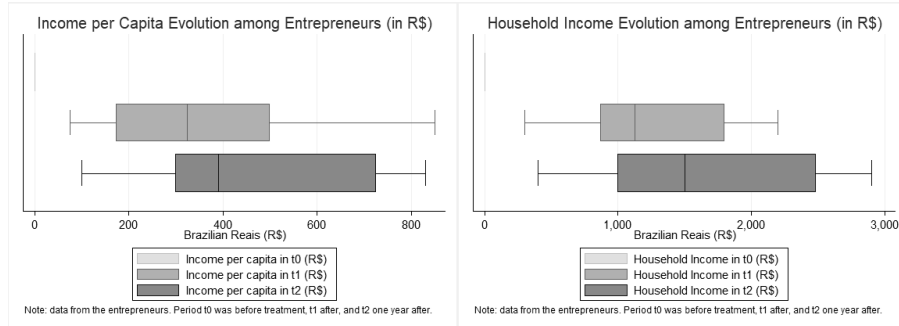
These descriptive results suggest that the effect of the program on dynamizing the favela market might be even higher than initially thought. This happens mainly by empowering local entrepreneurs to act as a catalyzer of market activities in the location.

Social Media Marketing from Entrepreneurs

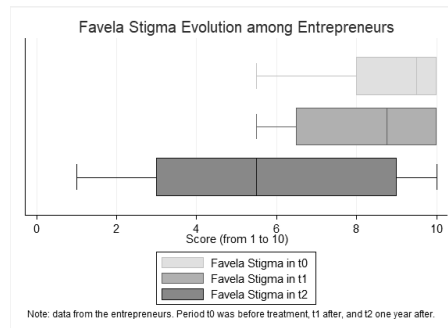
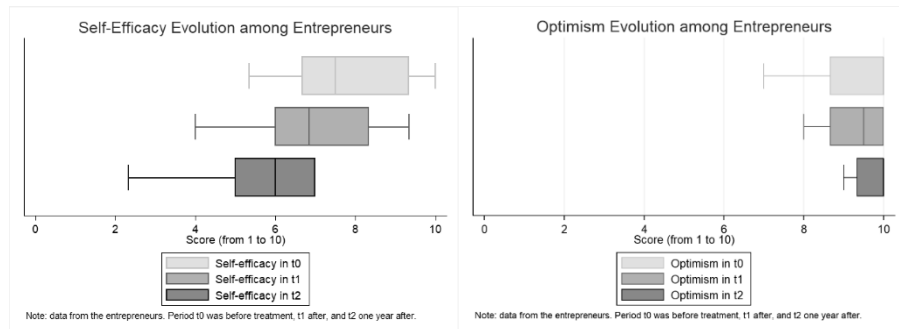


Source: Instagram® English translation. Both entrepreneurs authorized me to use their Instagram® pages here.

Income Evolution among Entrepreneurs



Socio-psychological Evolution among Entrepreneurs



Note: Raw data representation. Light gray represents baseline (survey t₀), medium gray represents endline (survey t₁), and dark gray one year after endline (survey t₂).

APPENDIX 12 – Additional Tests on Aggregated and Disaggregated Favela Stigma in t_1 on 25th, 50th, and 75th Quantiles Income per capita in t_1 (R\$)

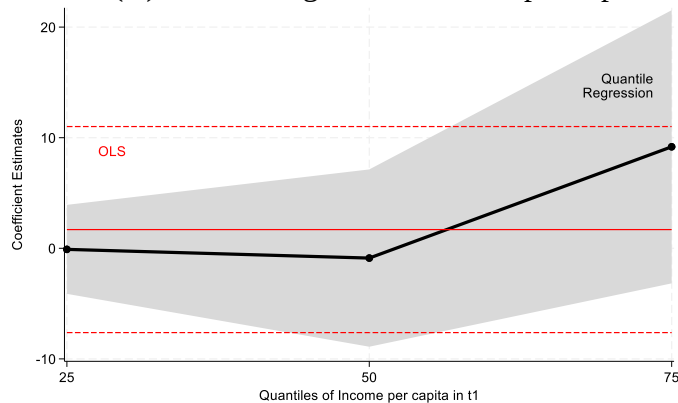
I further test a quantile regression model considering three specific quantiles of income per capita in t_1 (in R\$) distribution, respectively 25th, 50th, and 75th quantiles. This analysis enables me to evaluate how stigma operates conditionally on different quantiles of income per capita in t_1 (R\$). The general pattern we observe with deciles is also observed in this estimation with changes in significance in the upper quantile (75th) for the Favela Stigma measure ($p = 0.144$), and a significant effect in the upper quantile (75th) for AAAA ($p = 0.034$).

DV =	(0) OLS	(1) Income per capita in t_1 (R\$) q_{25}	(2) q_{50}	(3) q_{75}
Panel (A): Favela Stigma in t_1	1.693	-0.0954	-0.884	9.178
	(4.641)	(2.039)	(4.063)	(6.260)
Constant	175.6+	17.31	101.0	176.2
	(94.64)	(62.67)	(91.52)	(134.5)
Panel (B): “I often suffer from prejudice because I live in a favela” in t_1	3.622	-0.0642	3.314	10.15*
	(4.087)	(1.975)	(3.392)	(4.762)
Constant	166.8+	17.02	76.68	163.6
	(94.24)	(59.11)	(93.60)	(138.9)
Panel (C): “I believe I have a harder time getting a job because I live in a favela” in t_1	-1.267	-0.0792	-4.049	3.956
	(3.904)	(1.743)	(3.662)	(5.756)
Constant	191.8*	0	117.1	181.5
	(95.99)	(64.13)	(81.36)	(147.9)
Controls	Y	Y	Y	Y
DV Average by Quantile (in R\$)	214	10.47	158.78	477.61
Number of Individuals in Each Quantile	207	69	70	68
Observations	207	207	207	207
Bootstrap Replications	N/A	500	500	500

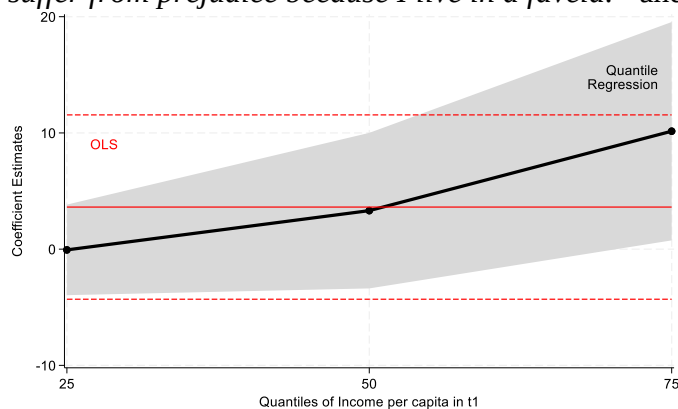
Note: Controls include individual-level characteristics, namely: job allocation at the endline (survey t_1), income level at the baseline (survey t_0), and demographic characteristics at the baseline (survey t_0). Socio-psychological and personality trait baseline controls were not included to avoid “bad control” issues, given their strong intertemporal correlation. Models estimated using 500 bootstrap replications. First and second quantiles represent all individuals with zero income per capita in t_1 (53 in total). Q8, in Panel (C) had 206 respondents at the endline (survey t_1). Darker shades indicate higher scores. Robust standard errors in parentheses. ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$

Visual Analysis of Aggregated and Disaggregated Favela Stigma in t_1 on 25th, 50th, and 75th Quantiles Income per capita in t_1 (R\$)

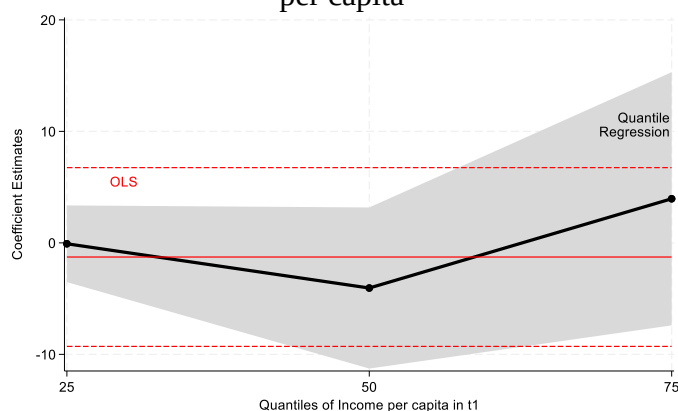
Panel (A): Favela Stigma and Income per capita



Panel (B): “I often suffer from prejudice because I live in a favela.” and Income per capita



Panel (C): “I believe I have a harder time getting a job because I live in a favela.” and Income per capita



Note: Point estimates (solid line) and 95% C.I. (shaded area and dashed line), in red dashed (OLS), in black shaded (Quantile Regression). All results are controlled by individual-level characteristics, namely: job allocation at the endline (survey t_1), income level at the baseline (survey t_0), and demographic characteristics at the baseline (survey t_0). Socio-psychological and personality trait baseline controls were not included to avoid “bad control” issues, given their strong intertemporal correlation. Q8, in Panel (C) had 206 respondents at the endline (survey t_1). Models estimated using 500 bootstrap replications.

APPENDIX 13 – Mediation Effect of Socio-Psychological Outcomes on Economic Outcomes

The improvements in the “socio-psychological” experiences of participants can manifest themselves not only as an improvement in well-being, but also as a catalyzer of individuals’ “technical” skills, and consequently, an improvement in their economic outcomes, particularly income gains. The study examines whether there is – there is not – a mediation effect of socio-psychological outcomes on the economic outcomes (Baron and Kenny, 1986; Iacobucci, Saldanha, and Deng, 2007; Zhao, Lynch, and Chen, 2010).

The OLS mediation test shows that self-efficacy has a partial mediation effect on the income per capita gains of participants (Baron and Kenny, 1986). Based on the results of models (1) and (2), gains in self-efficacy are responsible for 20 to 30 percent of the income gains ($p < 0.01$) by the end of the program. This result shows not only that the socio-psychological improvements benefit participants but that they also manifest themselves in monetary terms. Interestingly, optimism and social stigma do not have any statistical effect on income generation. Nevertheless, the coefficient signs of both variables are consistent with theoretical priors. These findings are consistent with the *business capabilities development* predictions about the program's effects.

The study presents additional tests using a Generalized Structural Equation Model (GSEM). Specialized literature on mediation effects argues that simple OLS mediation tests might generate biased estimates because they rely on three separate regressions. GSEM is therefore preferred as it corrects measurement errors and provides a consistent and asymptotically normal estimate by employing a maximum likelihood estimator (Iacobucci, Saldanha, and Deng, 2007;

Zhao, Lynch, and Chen, 2010; Kaplan and Vakili, 2015).⁷ The study presents a GSEM using a Gaussian function for all regressions. Model (4) of the GSEM mediation shows that self-efficacy partially mediates (22 percent) the effect of the treatment on income per capita gains of participants at ($p < 0.01$). This result corroborates the findings from the traditional mediation test both in significance and in magnitude.⁸

OLS Mediation Effect of Socio-Psychological Aspects on Economic Outcomes

DVs =	(1) Income per capita in t ₁ (R\$)	(2) Income per capita in t ₁ (R\$)	(3) Income per capita in t ₁ (R\$)	(4) Income per capita in t ₁ (R\$)	(5) Income per capita in t ₁ (R\$)	(6) Income per capita in t ₁ (R\$)
Treatment	78.71** (29.28)	92.22** (33.51)	94.04** (29.96)	105.0** (34.88)	99.29** (30.09)	108.4** (34.69)
Self-Efficacy in t₁	32.97** (9.366)	26.91** (9.148)				
Optimism in t₁			16.44+ (8.653)	3.144 (11.12)		
Favela Stigma in t₁					-1.091 (5.355)	-2.843 (6.787)
Constant	-61.81 (67.95)	-12.91 (262.5)	17.93 (75.77)	49.41 (255.4)	168.1** (61.24)	72.90 (252.7)
Controls	N	Y	N	Y	N	Y
Observations	207	207	207	207	207	207
R-squared	0.122	0.264	0.071	0.239	0.080	0.238

Note: All results are controlled by randomization strata and training period. Additional controls include demographic characteristics, big five personality traits, and outcome values at the baseline (survey t₀). Robust standard errors in parentheses.

** $p < 0.01$, * $p < 0.05$, + $p < 0.1$

⁷ GSEM model considers the simultaneous relationship among the three prediction paths – *Path A*: direct effect of the treatment on the economic outcomes (income per capita), *Path B*: direct effect of the treatment on the socio-psychological outcomes, and *Path C*: Indirect effect of the socio-psychological outcomes in the economic outcomes (income per capita).

⁸ It is important to note that Model (1) – direct effect – has exactly the same values as Model (4) – total effect – given the orthogonality of treatment towards both income per capita gains and socio-psychological outcomes. In other words, this result is driven by the random assignment of participants to either treatment or control groups.

Generalized Structural Equation Model (GSEM) Mediation Effect of Socio-Psychological Aspects on Economic Outcomes

<i>DVs =</i>	(1) Direct Effect on Income per capita in t_1 (Paths A and C)	(2) Direct Effect on Socio- Psychological Outcomes in t_1 (Path B)	(3) Indirect Effect on Income per capita in t_1 (Paths B and C)	(4) Total Effect on Income per capita in t_1 ($A + [B \times C]$)
Treatment	96.21** (30.92)			96.21** (30.92)
Self-Efficacy in t_1	28.30** (8.793)	0.524** (0.196)	32.44** (9.210)	28.30** (8.793)
Optimism in t_1	-4.676 (10.69)	0.409* (0.172)	-0.517 (11.03)	-4.676 (10.69)
Favela Stigma in t_1	-3.782 (6.078)	0.742* (0.334)	-1.226 (6.190)	-3.782 (6.078)
Constant	15.99 (231.4)		10.46 (242.8)	15.99 (231.4)
Controls	Y	Y	Y	Y
Observations	207	207	207	207

Note: All results are controlled by randomization strata and training period. Additional controls include demographic characteristics, big five personality traits, and outcome values at the baseline (survey t_0). Robust standard errors in parentheses. ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$

APPENDIX 14 – Moderation Effect of Socio-Psychological Measures in t_1 on Income per capita in t_1 (R\$)

I further test whether the three main socio-psychological measures moderate the effect of the treatment on income per capita at t_1 . The logic would be similar to the mediation test, however, now the analysis check whether the socio-psychological variables attenuate or exacerbate the effect of the treatment. The results are mainly null. This might be due to lack of statistical power, or due to “bad control” issues (Angrist and Pischke, 2008). This later case seems reasonable since the socio-psychological variable as well as the income per capita are both outcomes caused by the treatment.

Moderation Test of Socio-Psychological variables on Income per capita in t_1

<i>DVs =</i>	(1) Income per capita in t_1 (R\$)	(2) Income per capita in t_1 (R\$)	(3) Income per capita in t_1 (R\$)	(4) Income per capita in t_1 (R\$)	(5) Income per capita in t_1 (R\$)	(6) Income per capita in t_1 (R\$)
Treatment	33.82 (116.2)	-34.39 (138.5)	-71.99 (140.8)	-29.89 (165.7)	104.9 (82.72)	193.0* (82.21)
Self-Efficacy in t_1	24.96* (11.65)	16.43 (12.92)				
Treatment $\times$ Self-Efficacy in t_1	6.648 (17.26)	17.67 (20.15)				
Optimism in t_1			6.857 (10.54)	-4.290 (12.84)		
Treatment $\times$ Optimism in t_1			18.24 (15.94)	14.68 (18.23)		
Favela Stigma in t_1					-1.469 (6.463)	3.234 (8.450)
Treatment $\times$ Favela Stigma in t_1					-0.612 (10.52)	-11.60 (10.28)
Constant	-12.01 (74.79)	37.97 (268.2)	98.09 (93.62)	107.9 (264.3)	169.7** (52.14)	53.90 (251.2)
Controls	N	Y	N	Y	N	Y
Observations	207	207	207	207	207	207
R-squared	0.086	0.267	0.057	0.240	0.045	0.243

Note: Controls include randomization strata, training period, demographic characteristics, big five personality traits, and outcome values at the baseline (survey t_0). Robust standard errors in parentheses. ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$

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